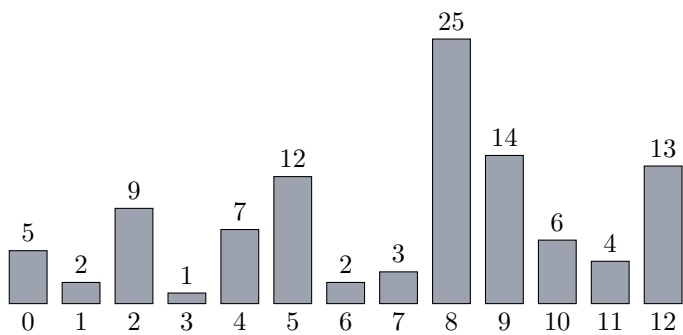


Quick Sort

Log

Chart

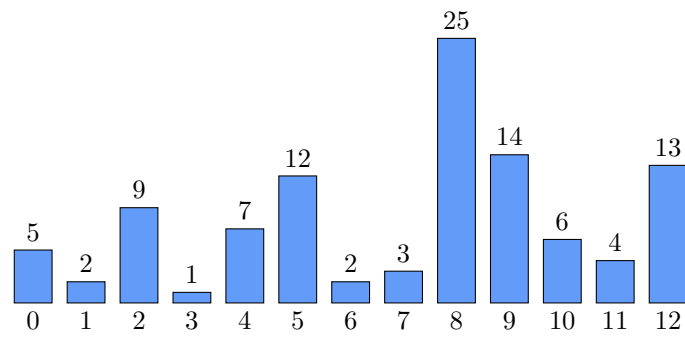


Quick Sort

Log

Partitioning and recursively sorting the section from index 0 to 12.

Chart

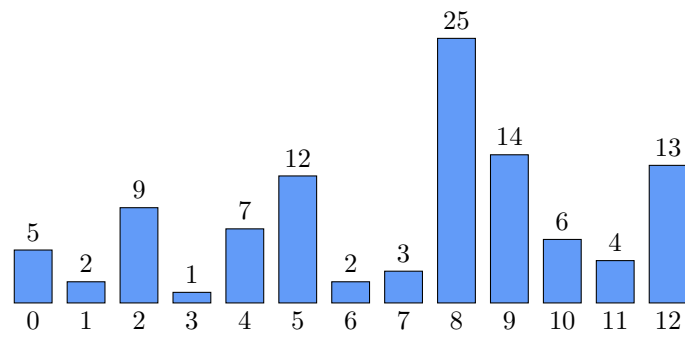


Quick Sort

Log

Using the last element of this section, value 13 at index 12, as the pivot.

Chart

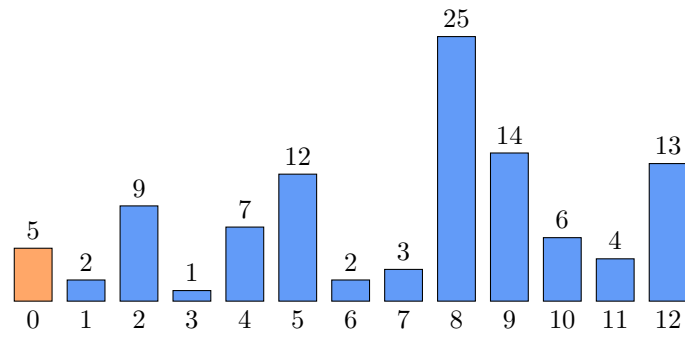


Quick Sort

Log

Checking whether the element at index 0 is smaller than the pivot (13).

Chart

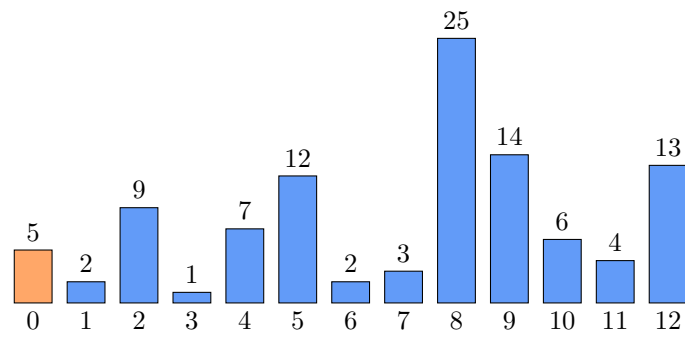


Quick Sort

Log

Element at index 0 is smaller than the pivot and already sits right after the smaller-than-pivot region

Chart

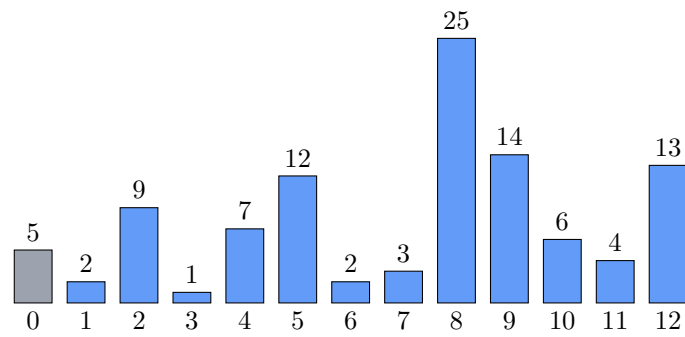


Quick Sort

Log

Element at index 0 is smaller than the pivot and already sits right after the smaller-than-pivot region

Chart

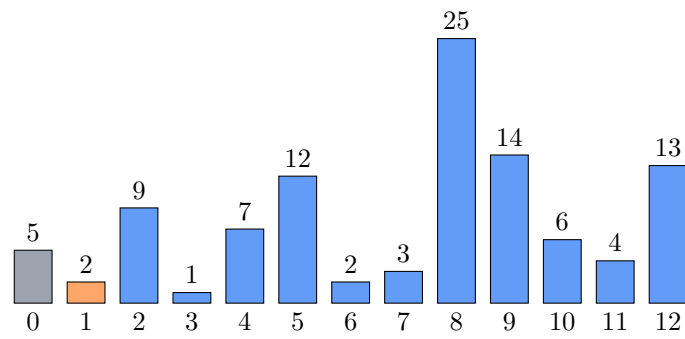


Quick Sort

Log

Checking whether the element at index 1 is smaller than the pivot (13).

Chart

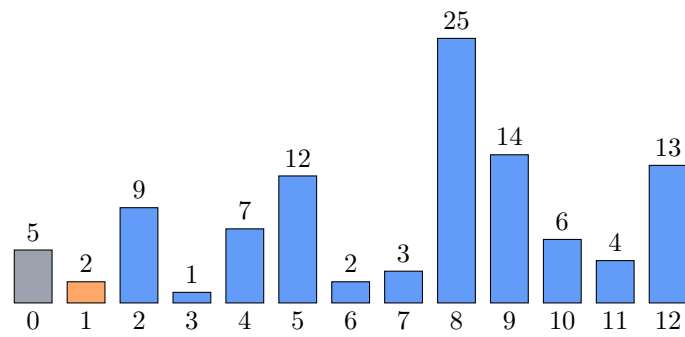


Quick Sort

Log

Element at index 1 is smaller than the pivot and already sits right after the smaller-than-pivot region

Chart

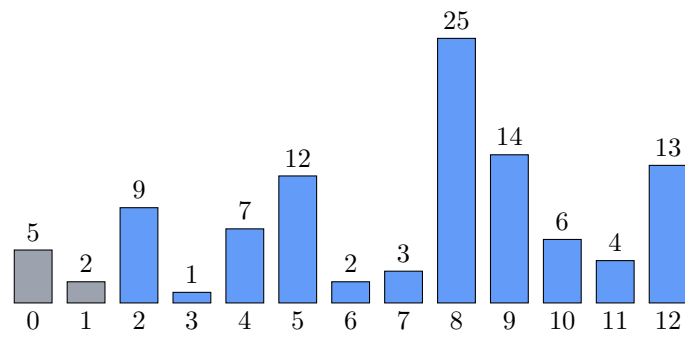


Quick Sort

Log

Element at index 1 is smaller than the pivot and already sits right after the smaller-than-pivot region

Chart

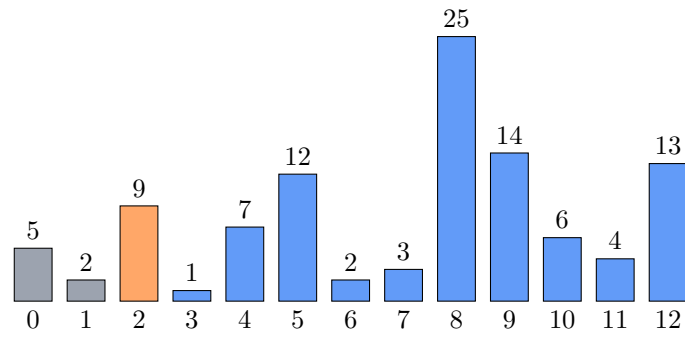


Quick Sort

Log

Checking whether the element at index 2 is smaller than the pivot (13).

Chart

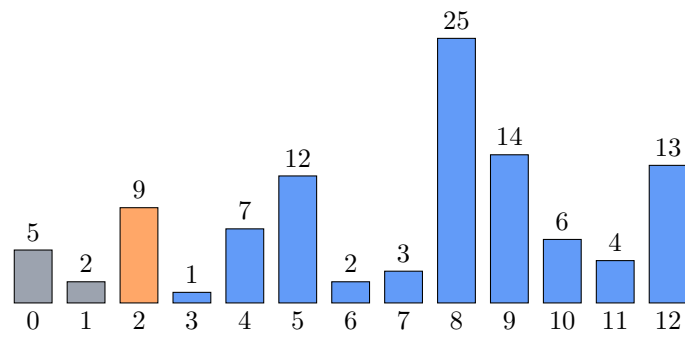


Quick Sort

Log

Element at index 2 is smaller than the pivot and already sits right after the smaller-than-pivot region

Chart

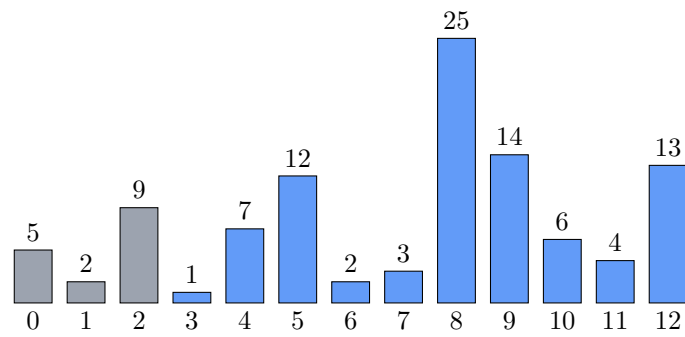


Quick Sort

Log

Element at index 2 is smaller than the pivot and already sits right after the smaller-than-pivot region

Chart

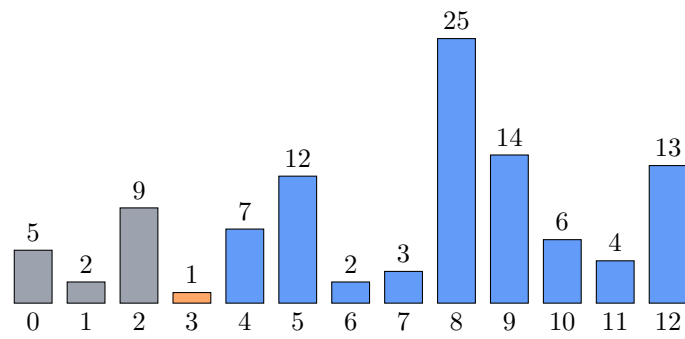


Quick Sort

Log

Checking whether the element at index 3 is smaller than the pivot (13).

Chart

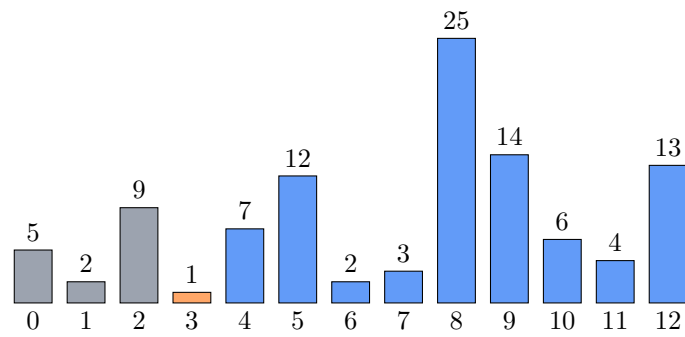


Quick Sort

Log

Element at index 3 is smaller than the pivot and already sits right after the smaller-than-pivot region

Chart

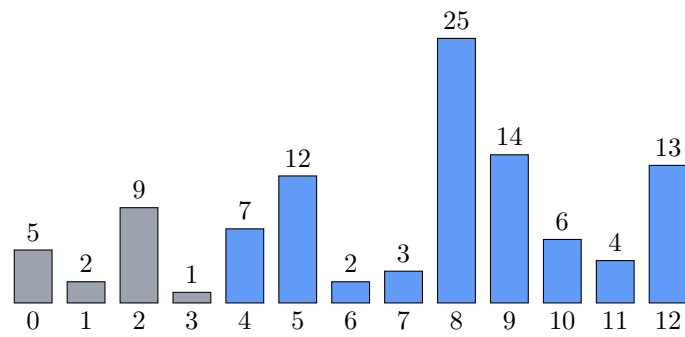


Quick Sort

Log

Element at index 3 is smaller than the pivot and already sits right after the smaller-than-pivot region

Chart

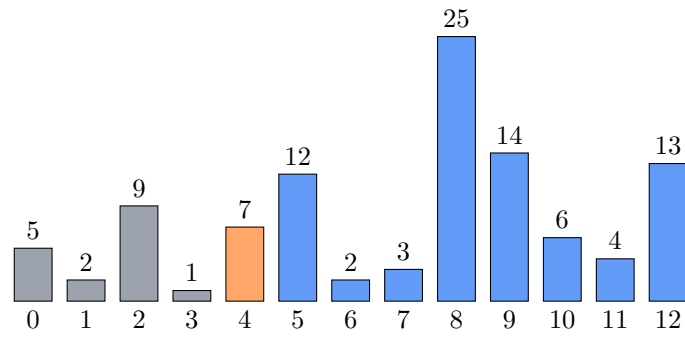


Quick Sort

Log

Checking whether the element at index 4 is smaller than the pivot (13).

Chart

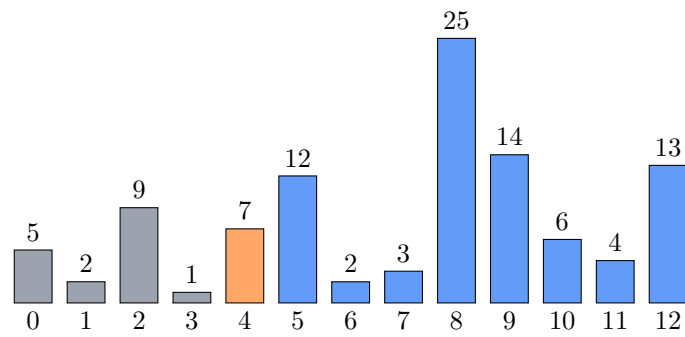


Quick Sort

Log

Element at index 4 is smaller than the pivot and already sits right after the smaller-than-pivot region

Chart

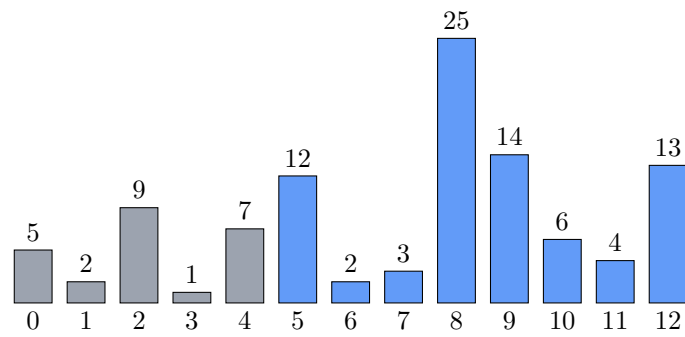


Quick Sort

Log

Element at index 4 is smaller than the pivot and already sits right after the smaller-than-pivot region

Chart

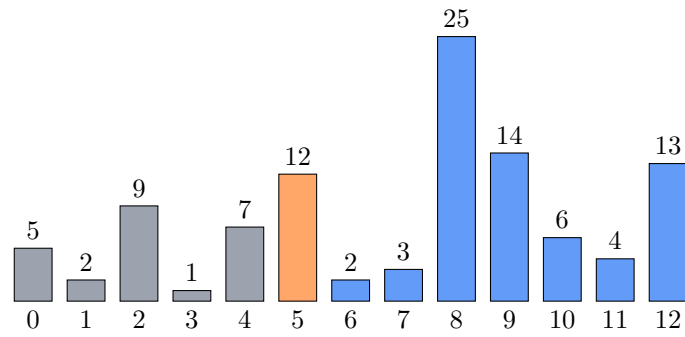


Quick Sort

Log

Checking whether the element at index 5 is smaller than the pivot (13).

Chart

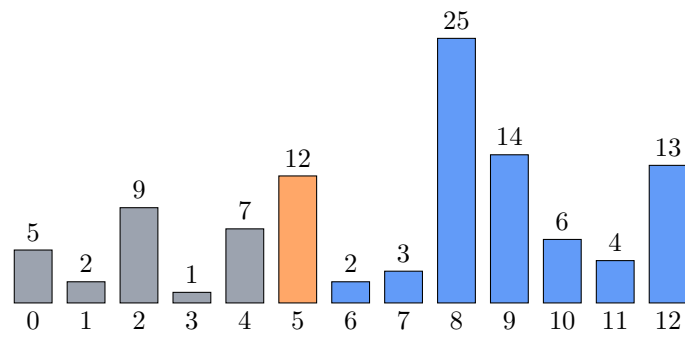


Quick Sort

Log

Element at index 5 is smaller than the pivot and already sits right after the smaller-than-pivot region

Chart

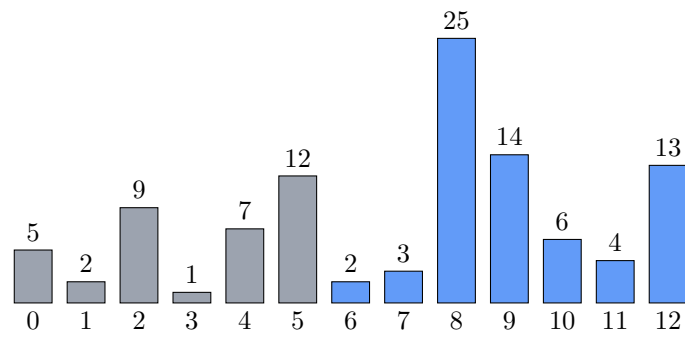


Quick Sort

Log

Element at index 5 is smaller than the pivot and already sits right after the smaller-than-pivot region

Chart

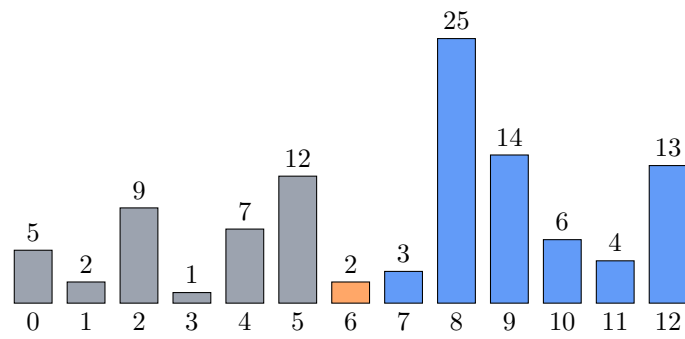


Quick Sort

Log

Checking whether the element at index 6 is smaller than the pivot (13).

Chart

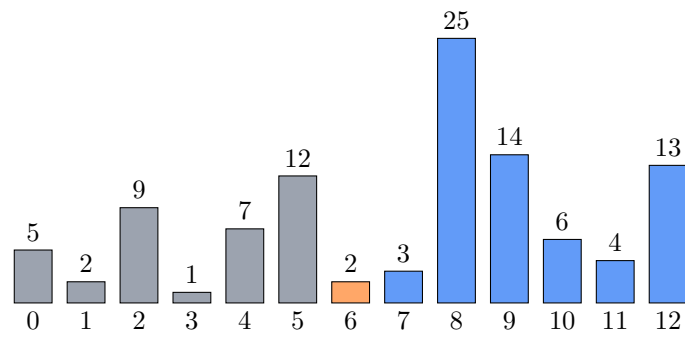


Quick Sort

Log

Element at index 6 is smaller than the pivot and already sits right after the smaller-than-pivot region

Chart

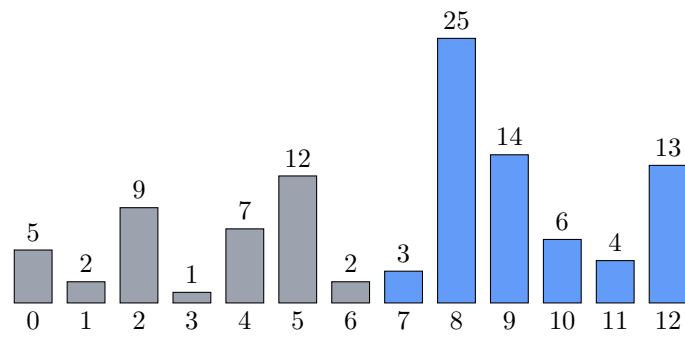


Quick Sort

Log

Element at index 6 is smaller than the pivot and already sits right after the smaller-than-pivot region

Chart

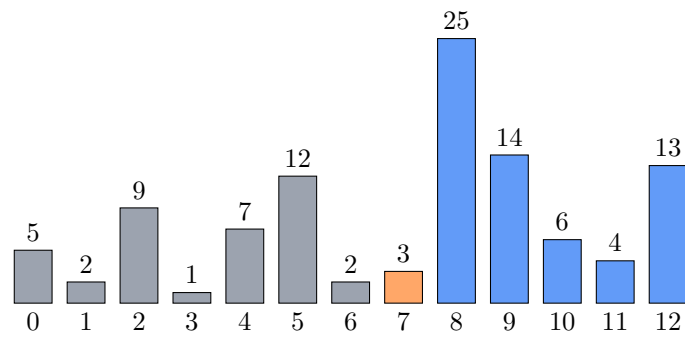


Quick Sort

Log

Checking whether the element at index 7 is smaller than the pivot (13).

Chart

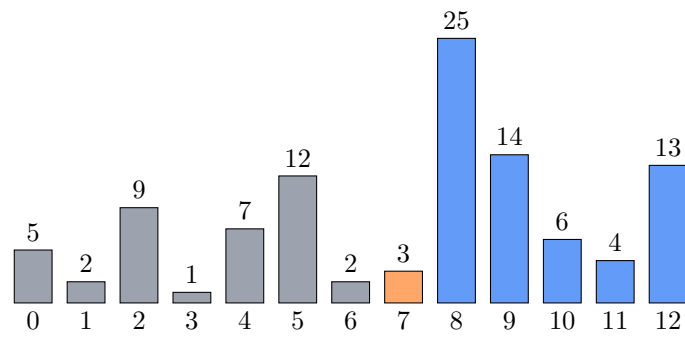


Quick Sort

Log

Element at index 7 is smaller than the pivot and already sits right after the smaller-than-pivot region

Chart

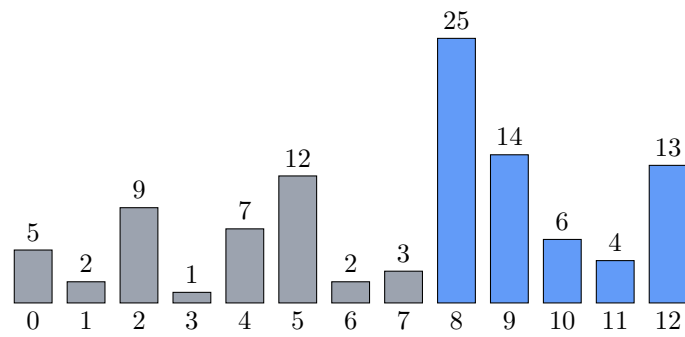


Quick Sort

Log

Element at index 7 is smaller than the pivot and already sits right after the smaller-than-pivot region

Chart

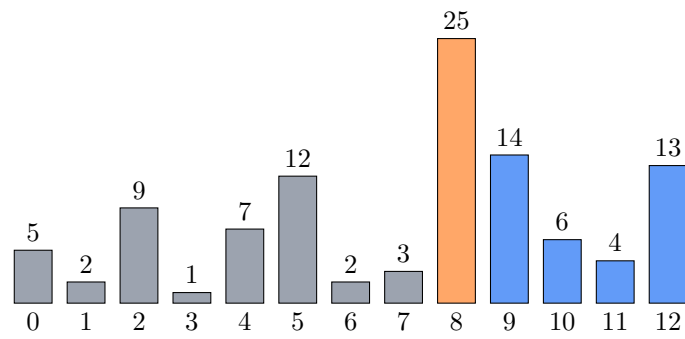


Quick Sort

Log

Checking whether the element at index 8 is smaller than the pivot (13).

Chart

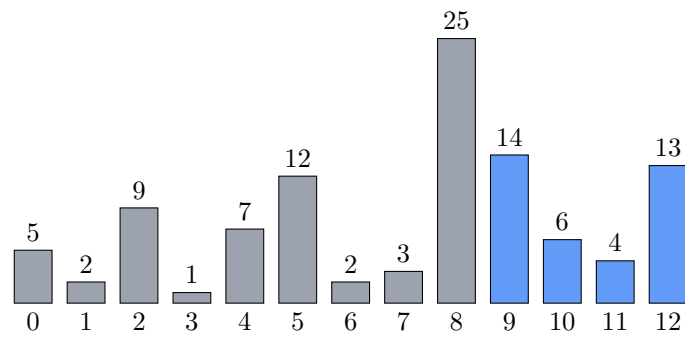


Quick Sort

Log

Checking whether the element at index 8 is smaller than the pivot (13).

Chart

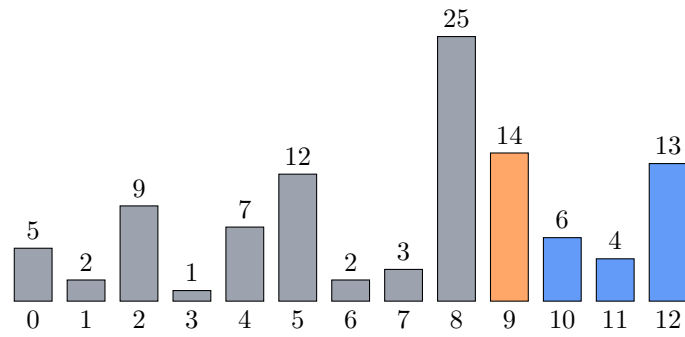


Quick Sort

Log

Checking whether the element at index 9 is smaller than the pivot (13).

Chart

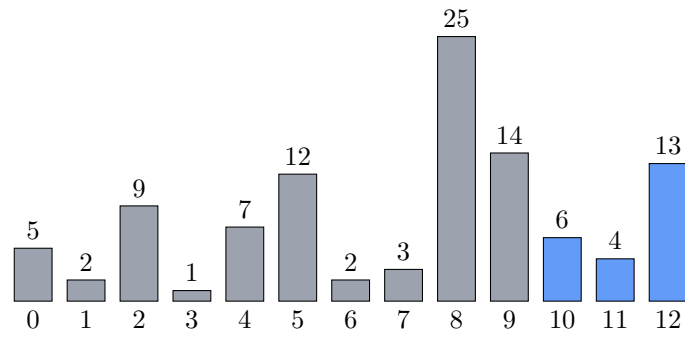


Quick Sort

Log

Checking whether the element at index 9 is smaller than the pivot (13).

Chart

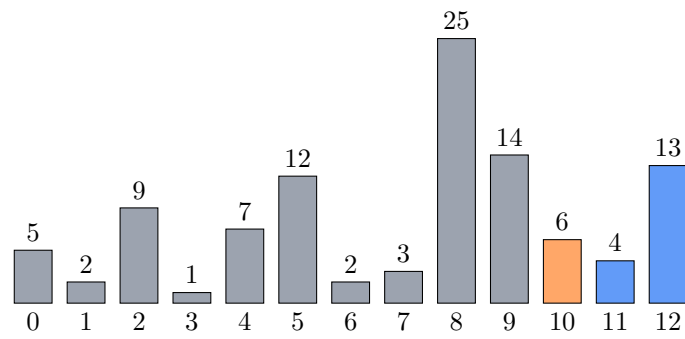


Quick Sort

Log

Checking whether the element at index 10 is smaller than the pivot (13).

Chart

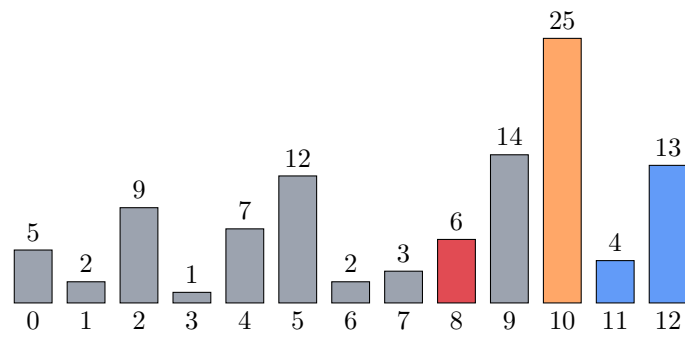


Quick Sort

Log

Element at index 10 is smaller than the pivot, so it swaps with index 8 to extend the region of values

Chart

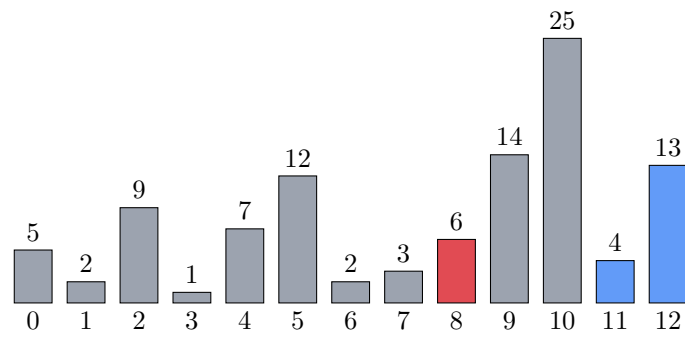


Quick Sort

Log

Element at index 10 is smaller than the pivot, so it swaps with index 8 to extend the region of values

Chart

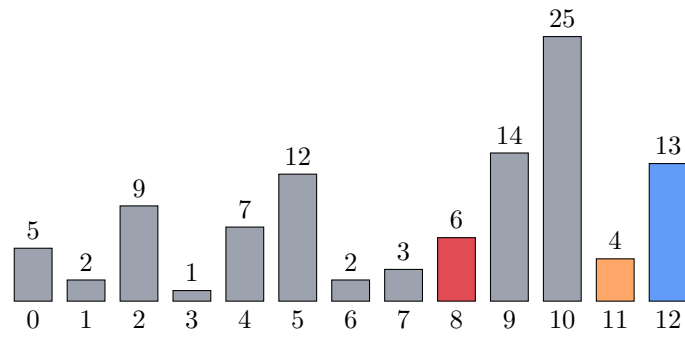


Quick Sort

Log

Checking whether the element at index 11 is smaller than the pivot (13).

Chart

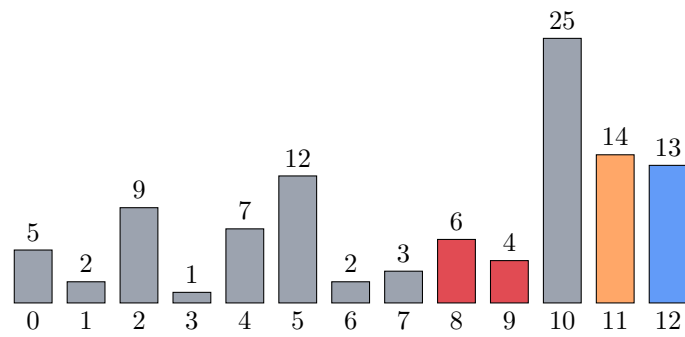


Quick Sort

Log

Element at index 11 is smaller than the pivot, so it swaps with index 9 to extend the region of values

Chart

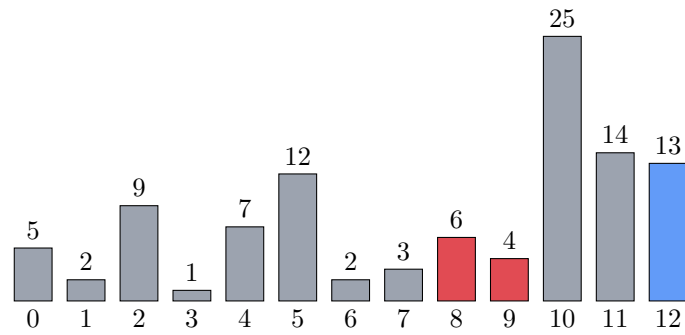


Quick Sort

Log

Element at index 11 is smaller than the pivot, so it swaps with index 9 to extend the region of values

Chart

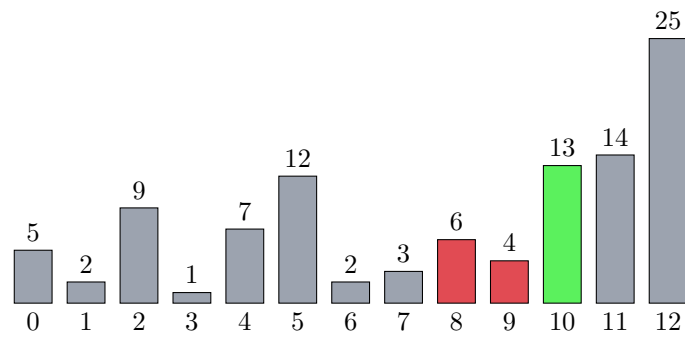


Quick Sort

Log

Swapping the pivot into index 10 | every element before it is now smaller, every element from there on

Chart

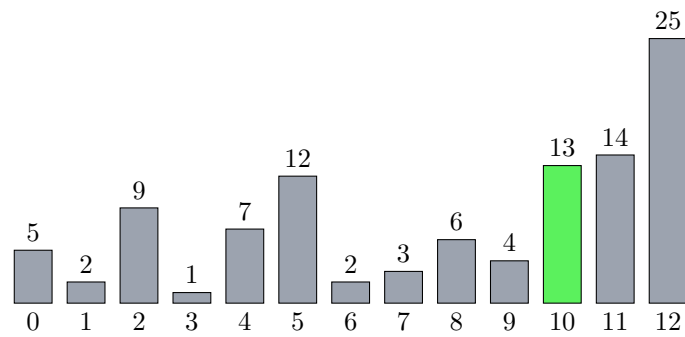


Quick Sort

Log

Swapping the pivot into index 10 | every element before it is now smaller, every element from there on

Chart

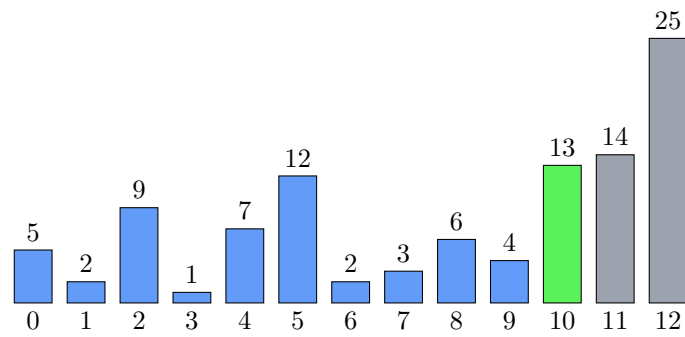


Quick Sort

Log

Partitioning and recursively sorting the section from index 0 to 9.

Chart

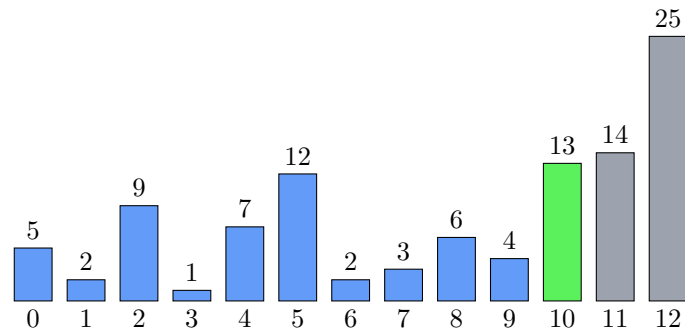


Quick Sort

Log

Using the last element of this section, value 4 at index 9, as the pivot.

Chart

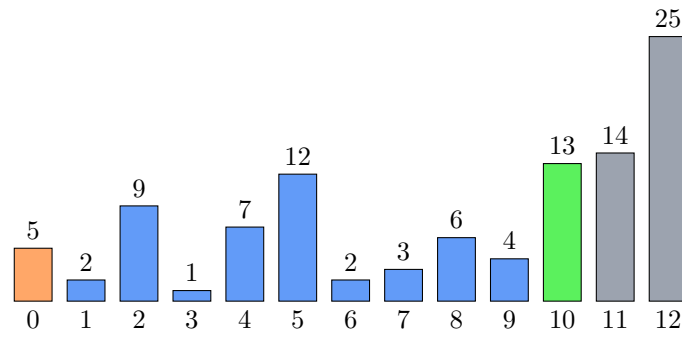


Quick Sort

Log

Checking whether the element at index 0 is smaller than the pivot (4).

Chart

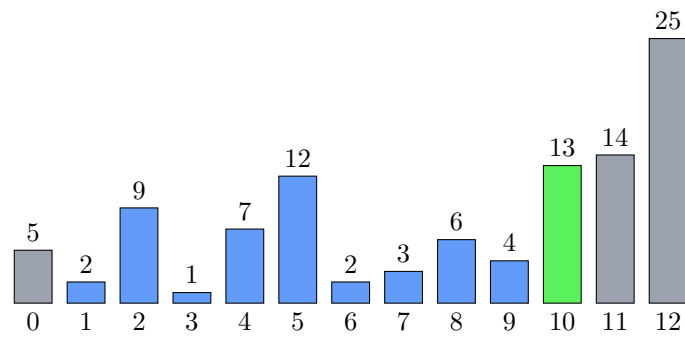


Quick Sort

Log

Checking whether the element at index 0 is smaller than the pivot (4).

Chart

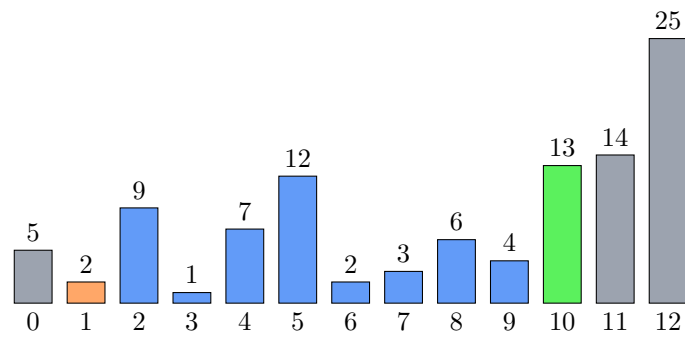


Quick Sort

Log

Checking whether the element at index 1 is smaller than the pivot (4).

Chart

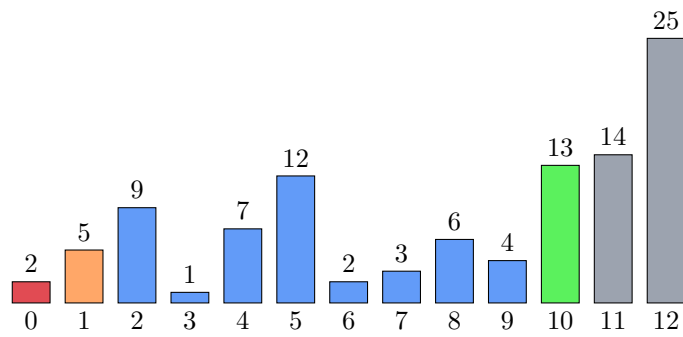


Quick Sort

Log

Element at index 1 is smaller than the pivot, so it swaps with index 0 to extend the region of values k

Chart

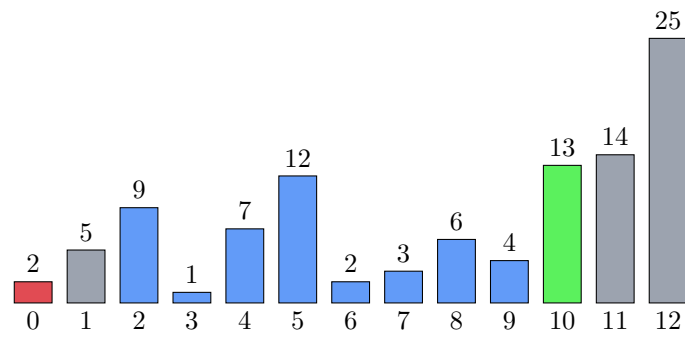


Quick Sort

Log

Element at index 1 is smaller than the pivot, so it swaps with index 0 to extend the region of values k

Chart

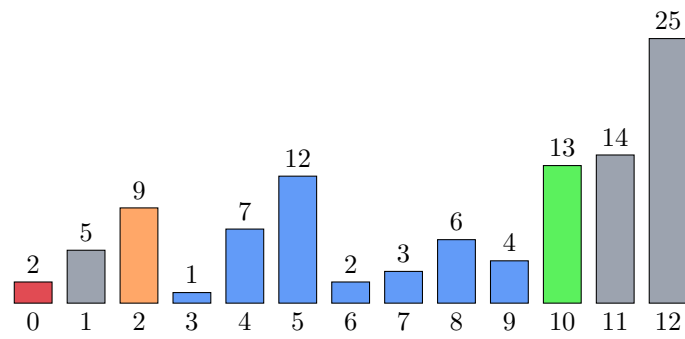


Quick Sort

Log

Checking whether the element at index 2 is smaller than the pivot (4).

Chart

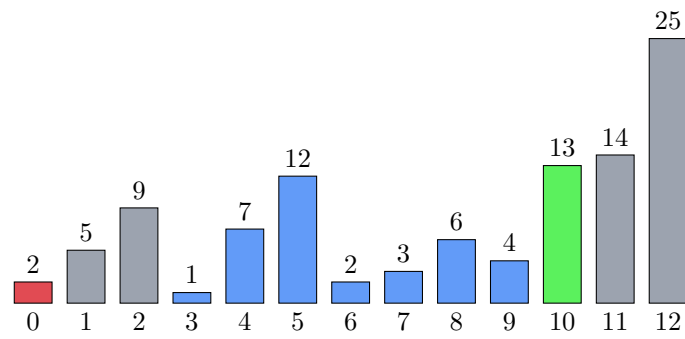


Quick Sort

Log

Checking whether the element at index 2 is smaller than the pivot (4).

Chart

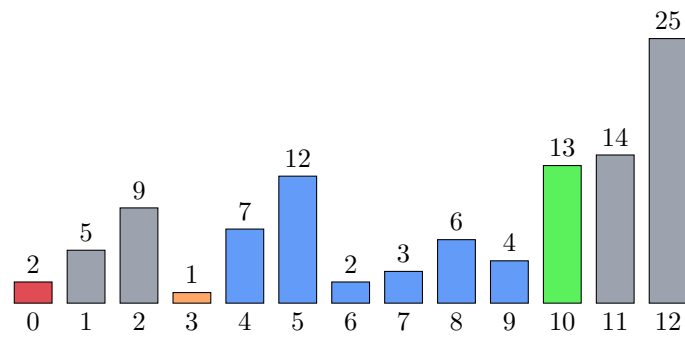


Quick Sort

Log

Checking whether the element at index 3 is smaller than the pivot (4).

Chart

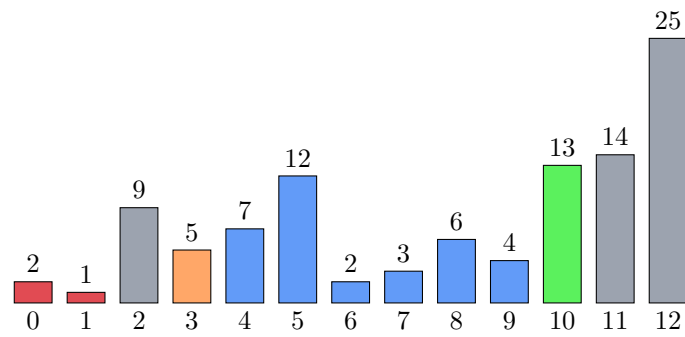


Quick Sort

Log

Element at index 3 is smaller than the pivot, so it swaps with index 1 to extend the region of values k

Chart

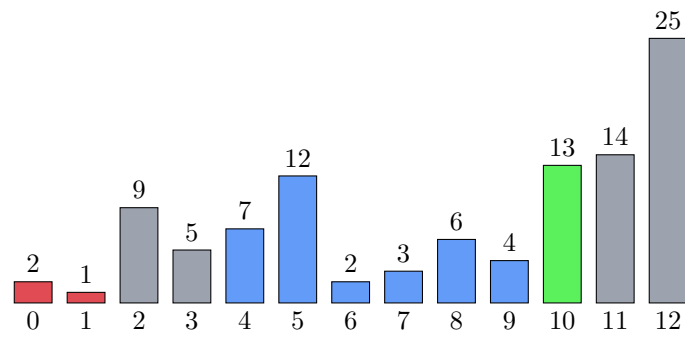


Quick Sort

Log

Element at index 3 is smaller than the pivot, so it swaps with index 1 to extend the region of values k

Chart

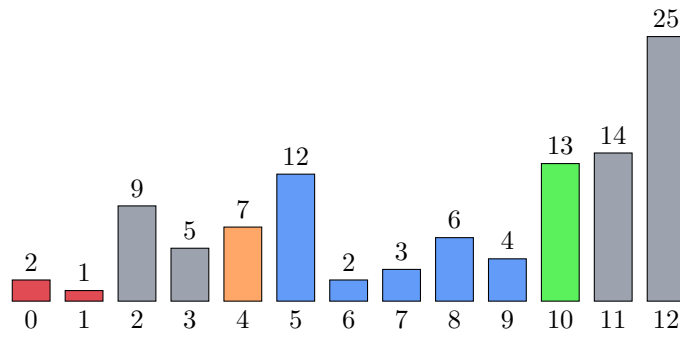


Quick Sort

Log

Checking whether the element at index 4 is smaller than the pivot (4).

Chart

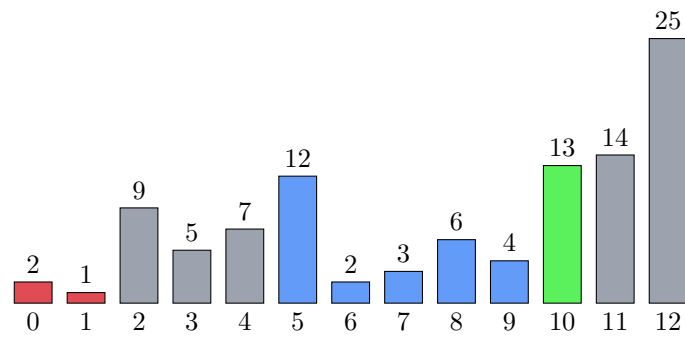


Quick Sort

Log

Checking whether the element at index 4 is smaller than the pivot (4).

Chart

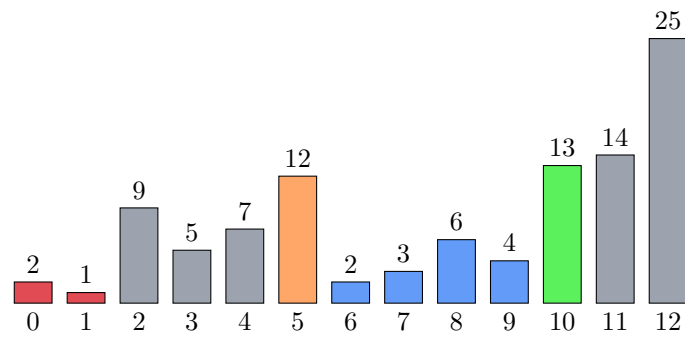


Quick Sort

Log

Checking whether the element at index 5 is smaller than the pivot (4).

Chart

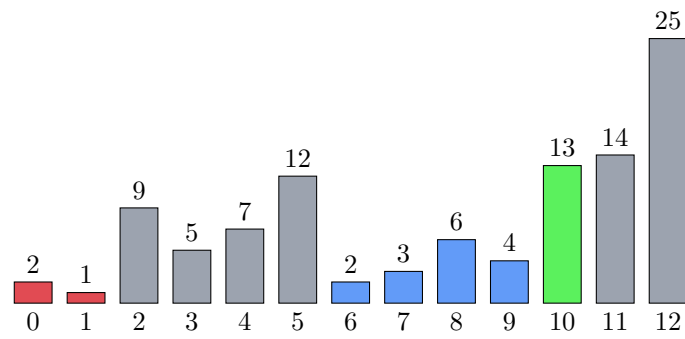


Quick Sort

Log

Checking whether the element at index 5 is smaller than the pivot (4).

Chart

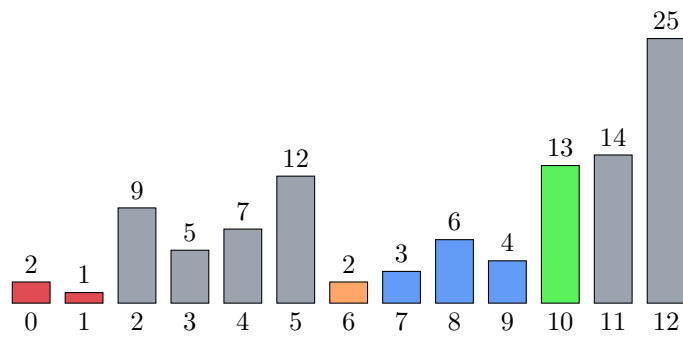


Quick Sort

Log

Checking whether the element at index 6 is smaller than the pivot (4).

Chart

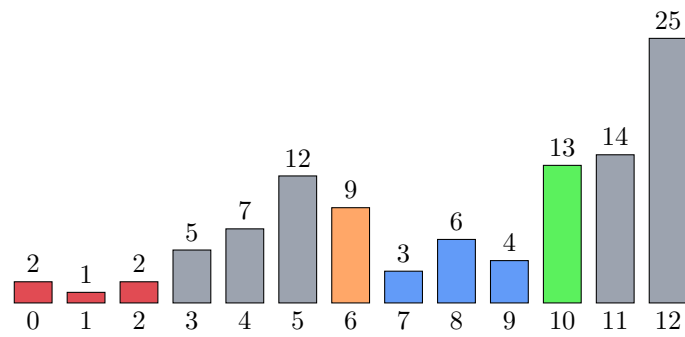


Quick Sort

Log

Element at index 6 is smaller than the pivot, so it swaps with index 2 to extend the region of values k

Chart

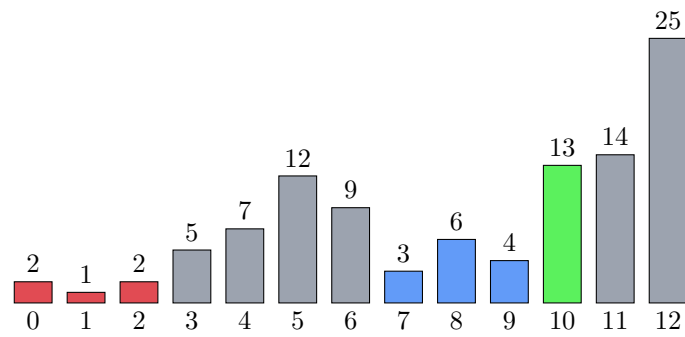


Quick Sort

Log

Element at index 6 is smaller than the pivot, so it swaps with index 2 to extend the region of values k

Chart

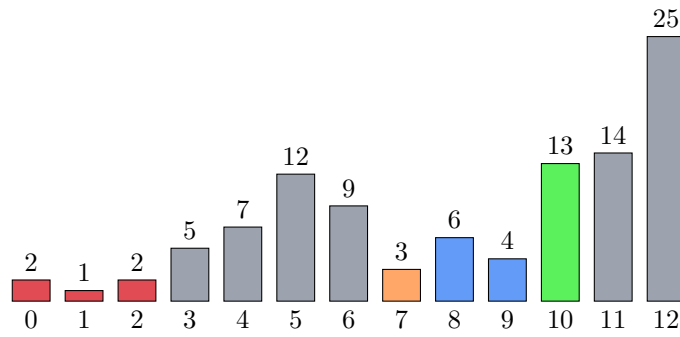


Quick Sort

Log

Checking whether the element at index 7 is smaller than the pivot (4).

Chart

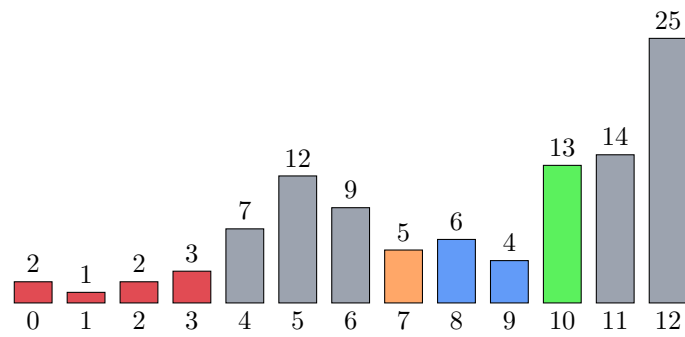


Quick Sort

Log

Element at index 7 is smaller than the pivot, so it swaps with index 3 to extend the region of values k

Chart

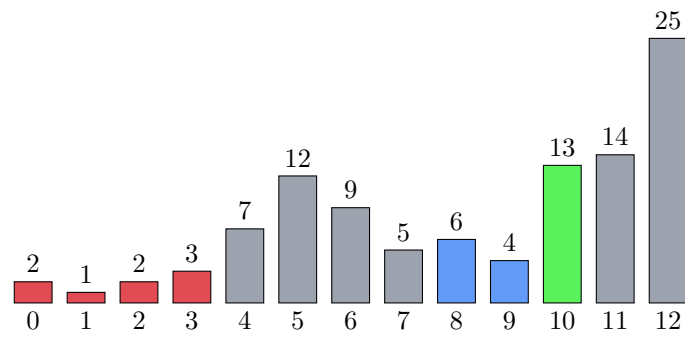


Quick Sort

Log

Element at index 7 is smaller than the pivot, so it swaps with index 3 to extend the region of values k

Chart

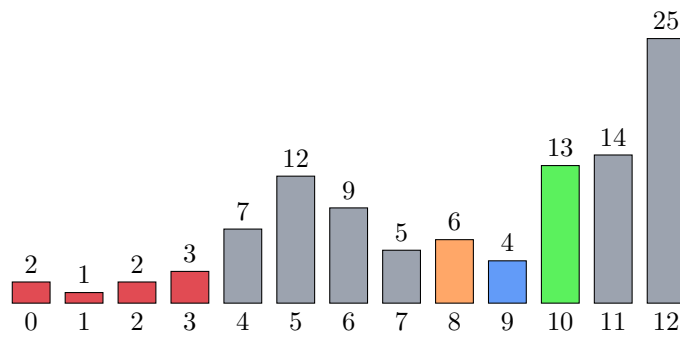


Quick Sort

Log

Checking whether the element at index 8 is smaller than the pivot (4).

Chart

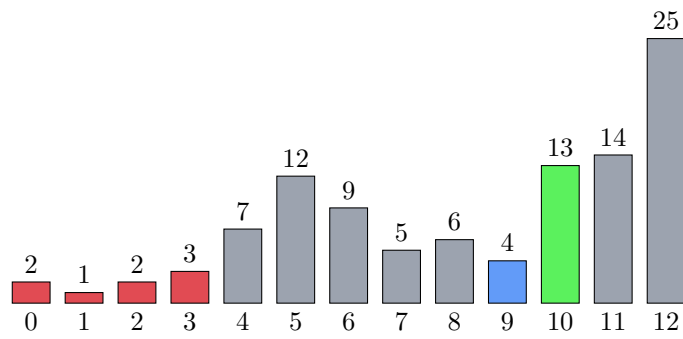


Quick Sort

Log

Checking whether the element at index 8 is smaller than the pivot (4).

Chart

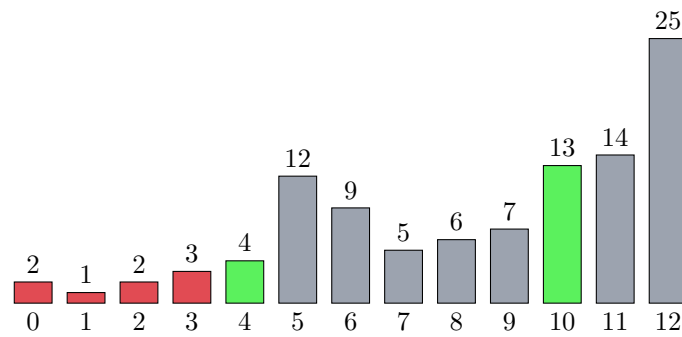


Quick Sort

Log

Swapping the pivot into index 4 | every element before it is now smaller, every element from there on i

Chart

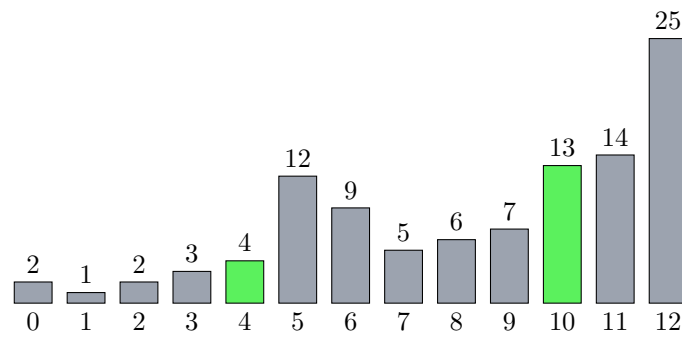


Quick Sort

Log

Swapping the pivot into index 4 | every element before it is now smaller, every element from there on i

Chart

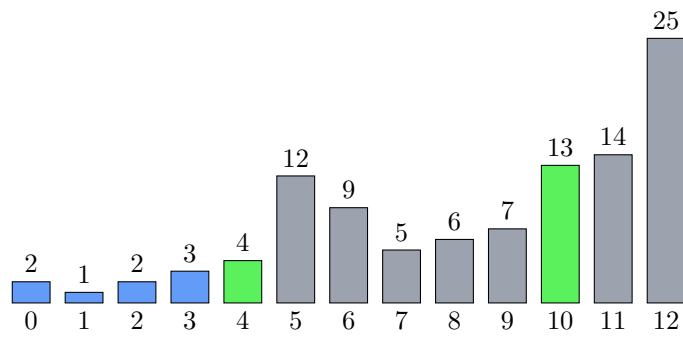


Quick Sort

Log

Partitioning and recursively sorting the section from index 0 to 3.

Chart

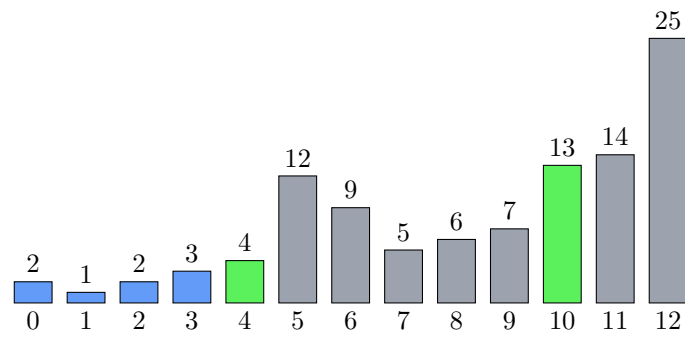


Quick Sort

Log

Using the last element of this section, value 3 at index 3, as the pivot.

Chart

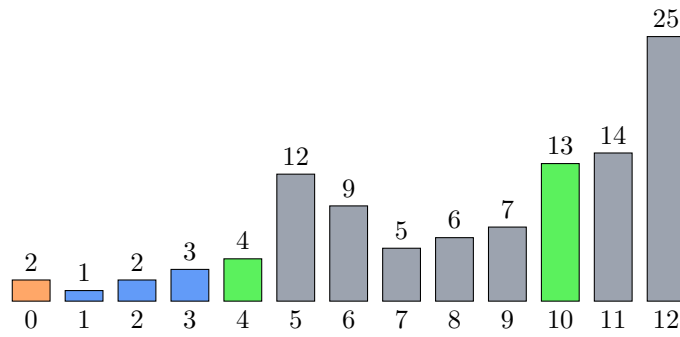


Quick Sort

Log

Checking whether the element at index 0 is smaller than the pivot (3).

Chart

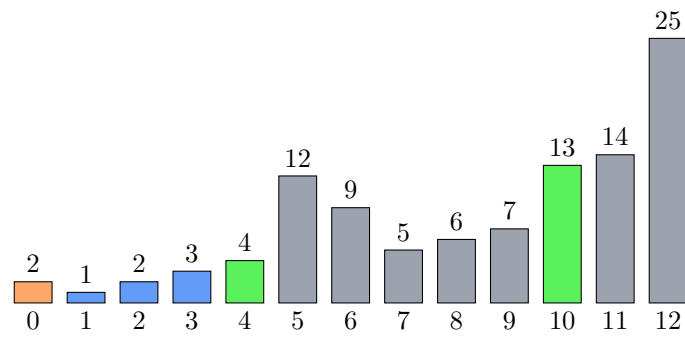


Quick Sort

Log

Element at index 0 is smaller than the pivot and already sits right after the smaller-than-pivot region

Chart

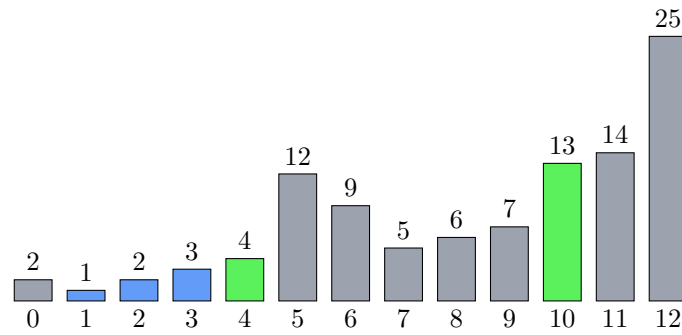


Quick Sort

Log

Element at index 0 is smaller than the pivot and already sits right after the smaller-than-pivot region

Chart

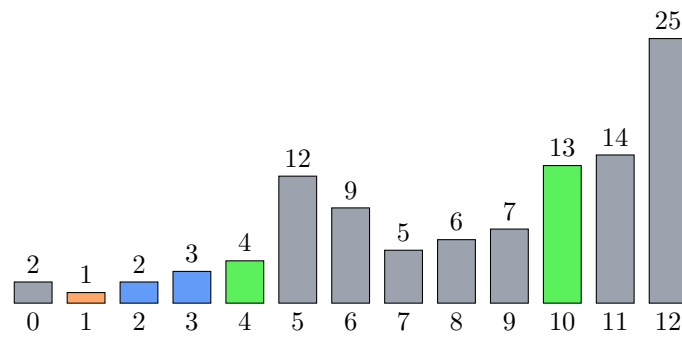


Quick Sort

Log

Checking whether the element at index 1 is smaller than the pivot (3).

Chart

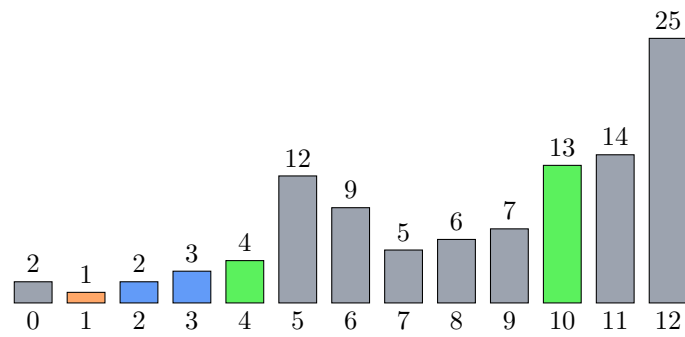


Quick Sort

Log

Element at index 1 is smaller than the pivot and already sits right after the smaller-than-pivot region

Chart

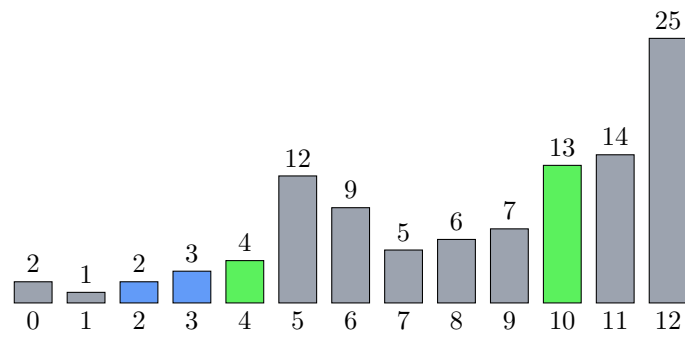


Quick Sort

Log

Element at index 1 is smaller than the pivot and already sits right after the smaller-than-pivot region

Chart

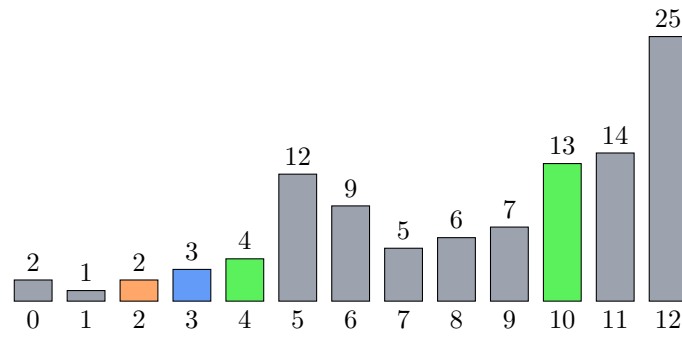


Quick Sort

Log

Checking whether the element at index 2 is smaller than the pivot (3).

Chart

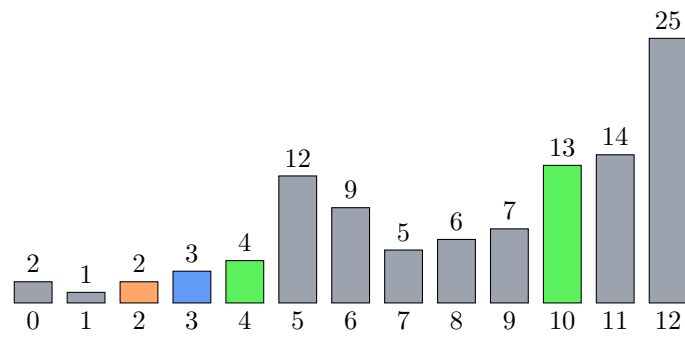


Quick Sort

Log

Element at index 2 is smaller than the pivot and already sits right after the smaller-than-pivot region

Chart

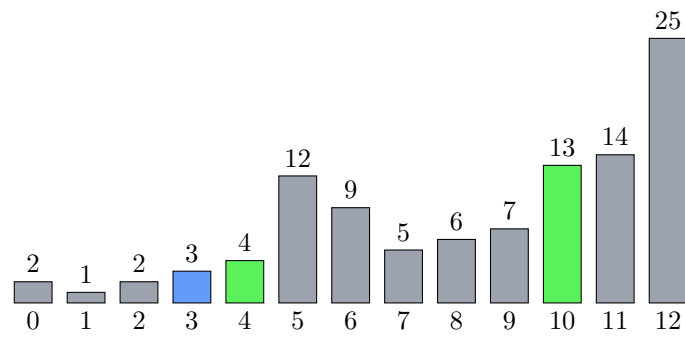


Quick Sort

Log

Element at index 2 is smaller than the pivot and already sits right after the smaller-than-pivot region

Chart

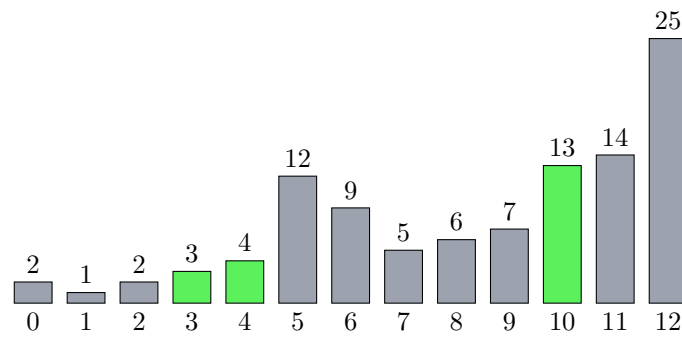


Quick Sort

Log

Swapping the pivot into index 3 | every element before it is now smaller, every element from there on i

Chart

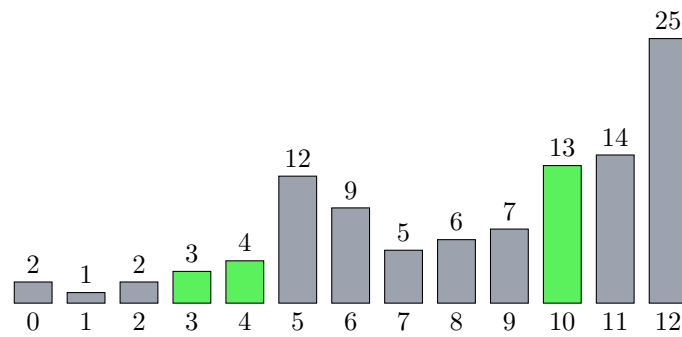


Quick Sort

Log

Swapping the pivot into index 3 | every element before it is now smaller, every element from there on i

Chart

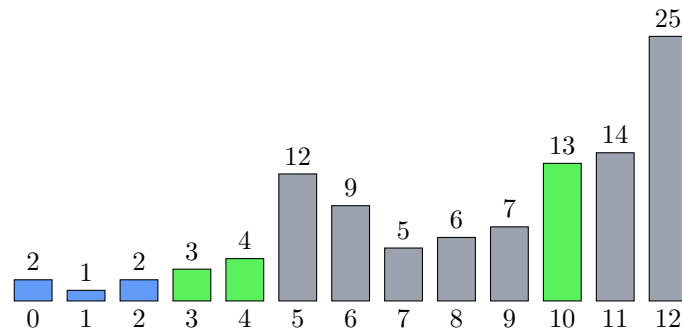


Quick Sort

Log

Partitioning and recursively sorting the section from index 0 to 2.

Chart

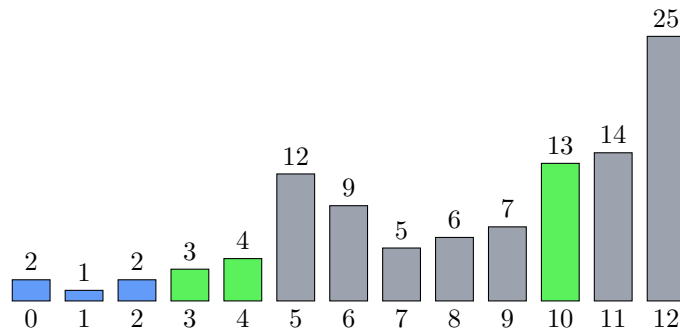


Quick Sort

Log

Using the last element of this section, value 2 at index 2, as the pivot.

Chart

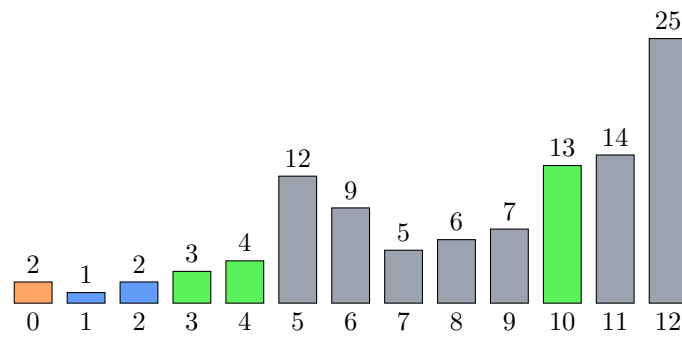


Quick Sort

Log

Checking whether the element at index 0 is smaller than the pivot (2).

Chart

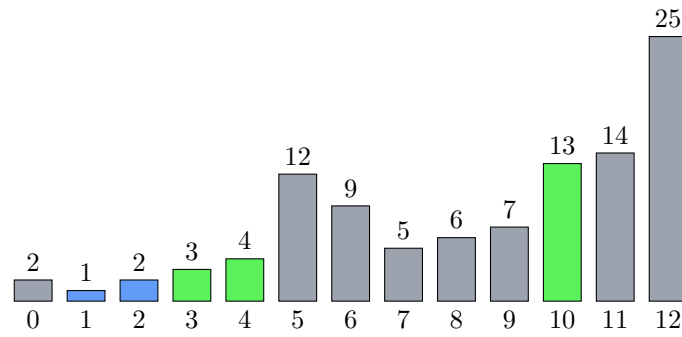


Quick Sort

Log

Checking whether the element at index 0 is smaller than the pivot (2).

Chart

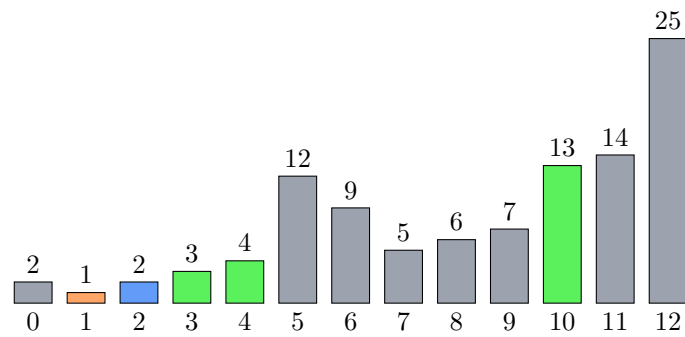


Quick Sort

Log

Checking whether the element at index 1 is smaller than the pivot (2).

Chart

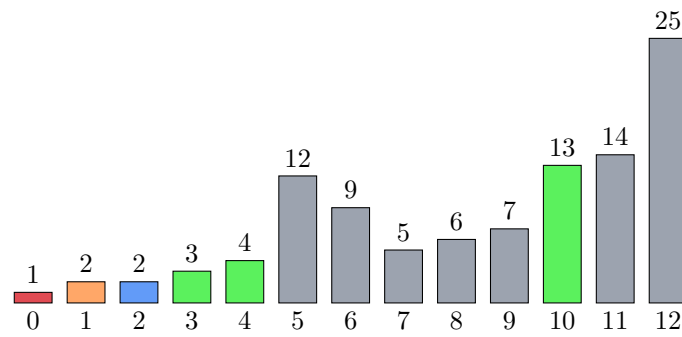


Quick Sort

Log

Element at index 1 is smaller than the pivot, so it swaps with index 0 to extend the region of values k

Chart

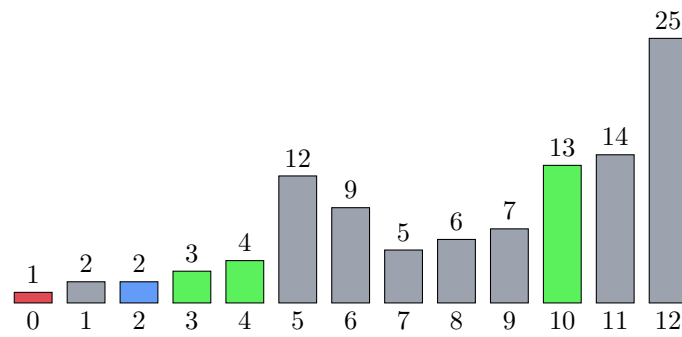


Quick Sort

Log

Element at index 1 is smaller than the pivot, so it swaps with index 0 to extend the region of values k

Chart

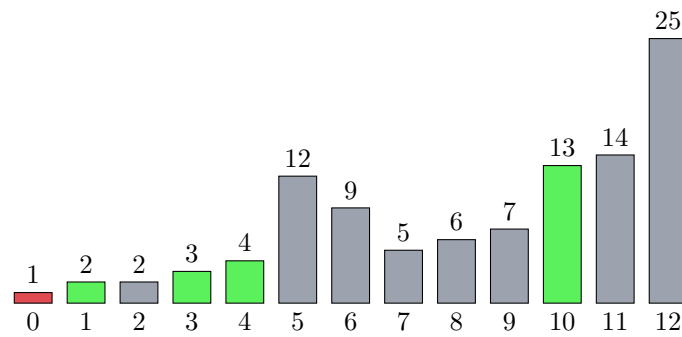


Quick Sort

Log

Swapping the pivot into index 1 | every element before it is now smaller, every element from there on i

Chart

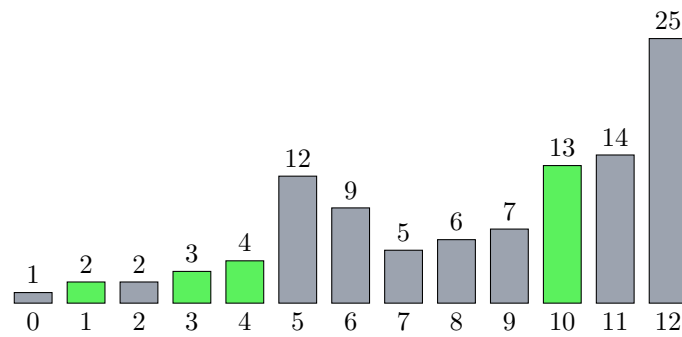


Quick Sort

Log

Swapping the pivot into index 1 | every element before it is now smaller, every element from there on i

Chart

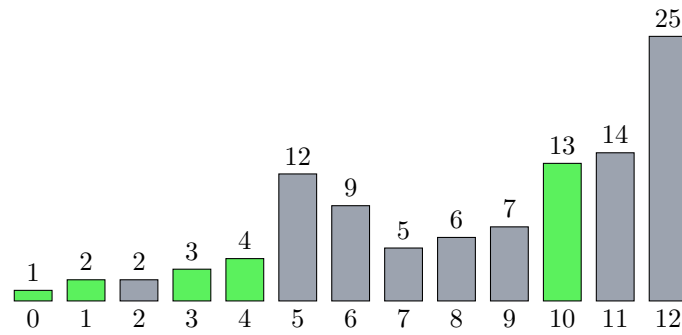


Quick Sort

Log

The section at index 0 contains only one element, so it's already in its final position.

Chart

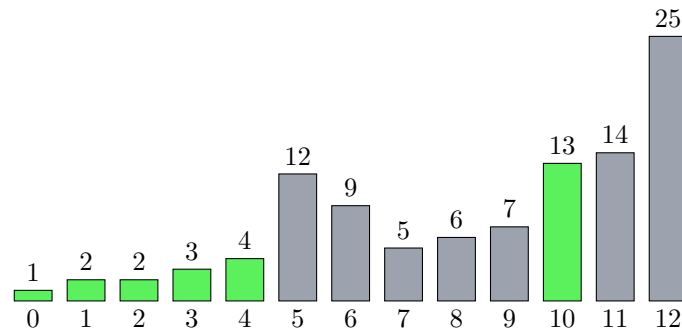


Quick Sort

Log

The section at index 2 contains only one element, so it's already in its final position.

Chart

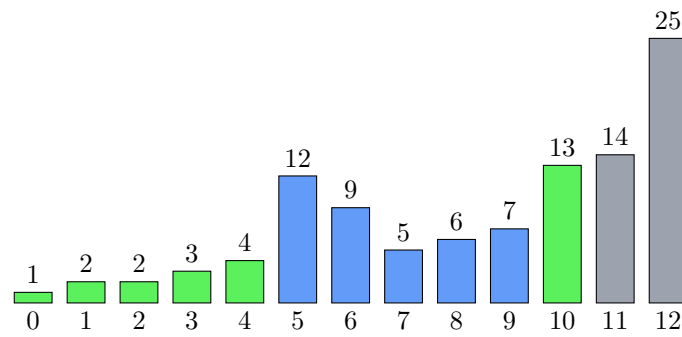


Quick Sort

Log

Partitioning and recursively sorting the section from index 5 to 9.

Chart

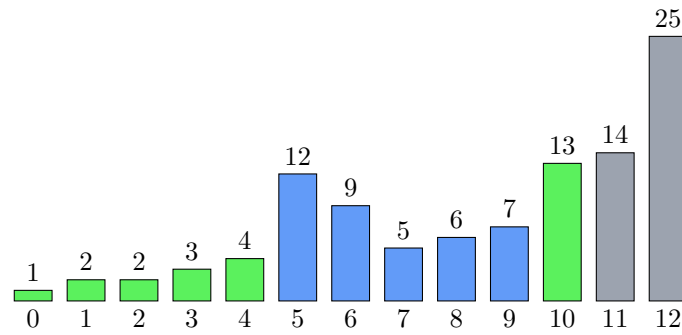


Quick Sort

Log

Using the last element of this section, value 7 at index 9, as the pivot.

Chart

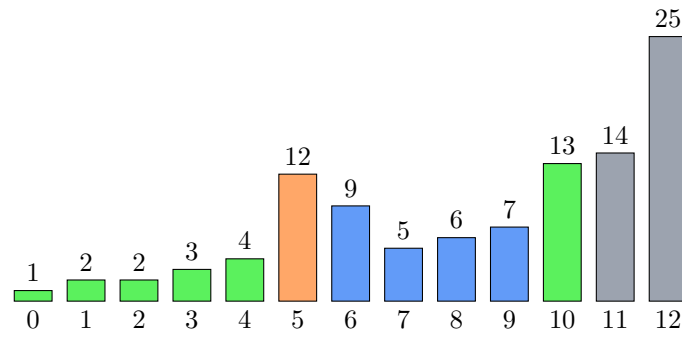


Quick Sort

Log

Checking whether the element at index 5 is smaller than the pivot (7).

Chart

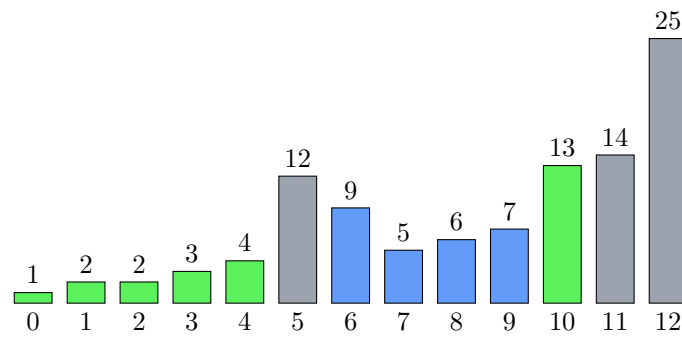


Quick Sort

Log

Checking whether the element at index 5 is smaller than the pivot (7).

Chart

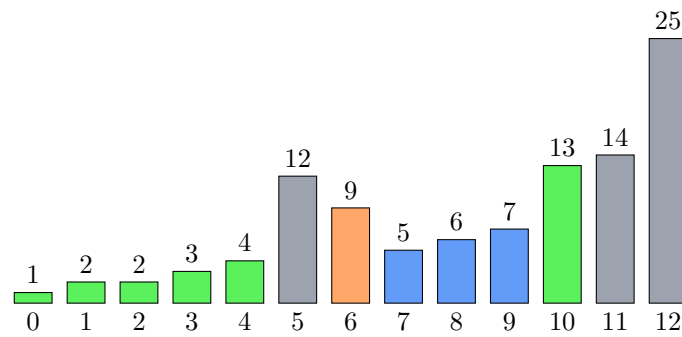


Quick Sort

Log

Checking whether the element at index 6 is smaller than the pivot (7).

Chart

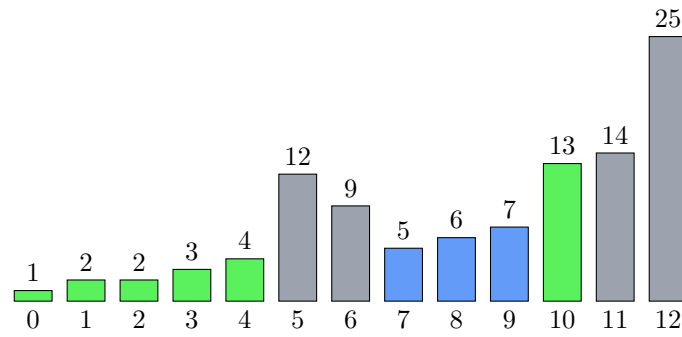


Quick Sort

Log

Checking whether the element at index 6 is smaller than the pivot (7).

Chart

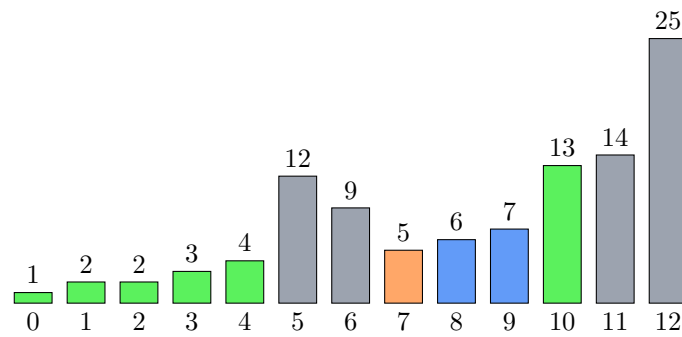


Quick Sort

Log

Checking whether the element at index 7 is smaller than the pivot (7).

Chart

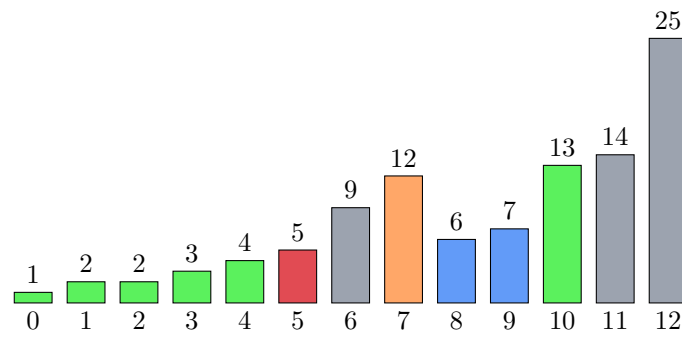


Quick Sort

Log

Element at index 7 is smaller than the pivot, so it swaps with index 5 to extend the region of values k

Chart

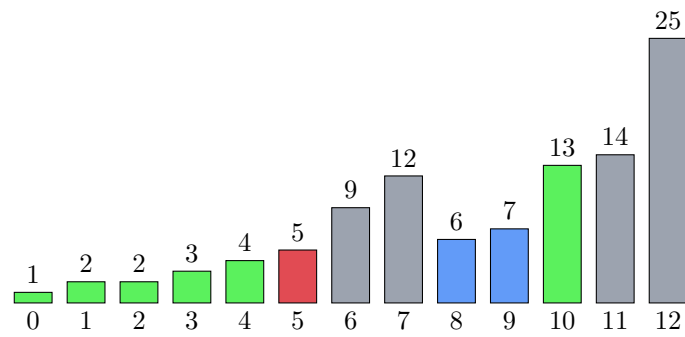


Quick Sort

Log

Element at index 7 is smaller than the pivot, so it swaps with index 5 to extend the region of values k

Chart

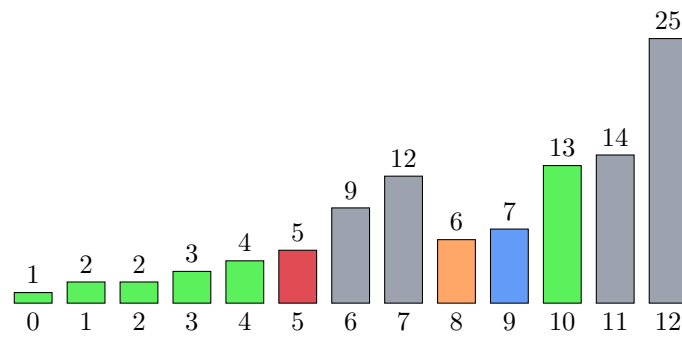


Quick Sort

Log

Checking whether the element at index 8 is smaller than the pivot (7).

Chart

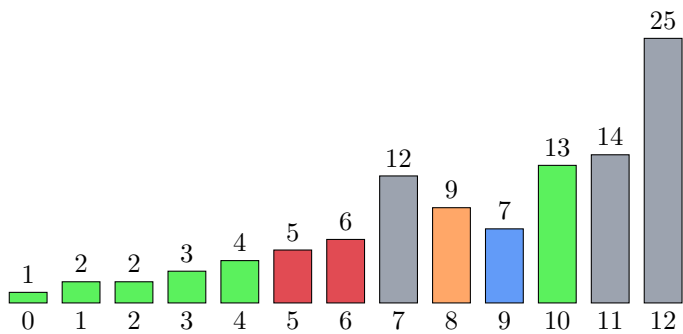


Quick Sort

Log

Element at index 8 is smaller than the pivot, so it swaps with index 6 to extend the region of values k

Chart

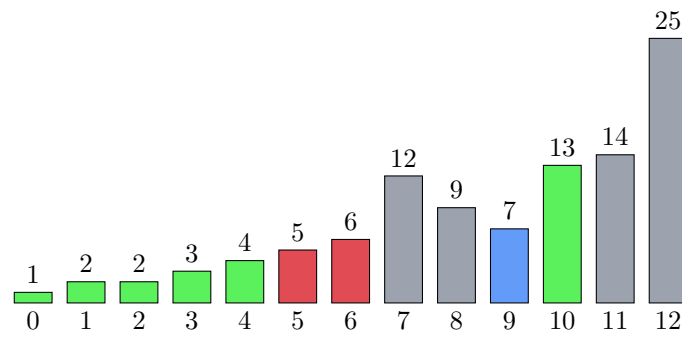


Quick Sort

Log

Element at index 8 is smaller than the pivot, so it swaps with index 6 to extend the region of values k

Chart

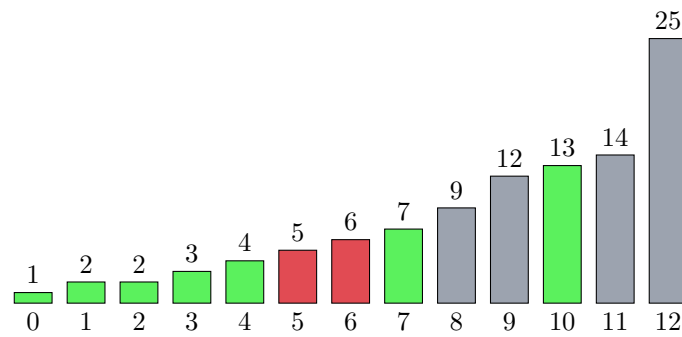


Quick Sort

Log

Swapping the pivot into index 7 | every element before it is now smaller, every element from there on i

Chart

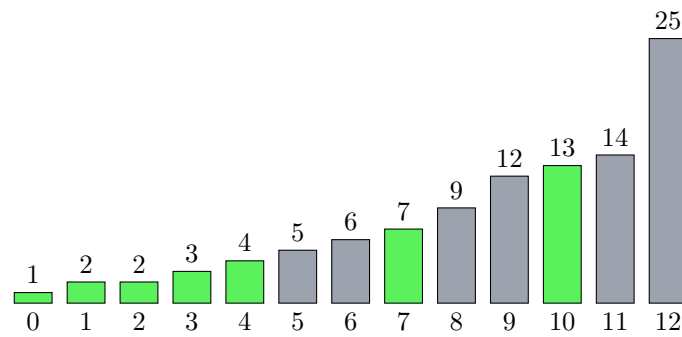


Quick Sort

Log

Swapping the pivot into index 7 | every element before it is now smaller, every element from there on i

Chart

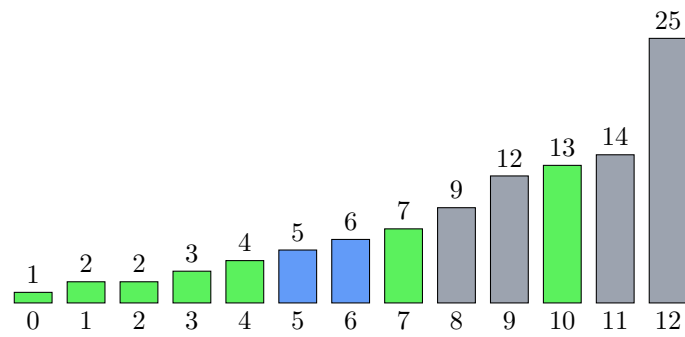


Quick Sort

Log

Partitioning and recursively sorting the section from index 5 to 6.

Chart

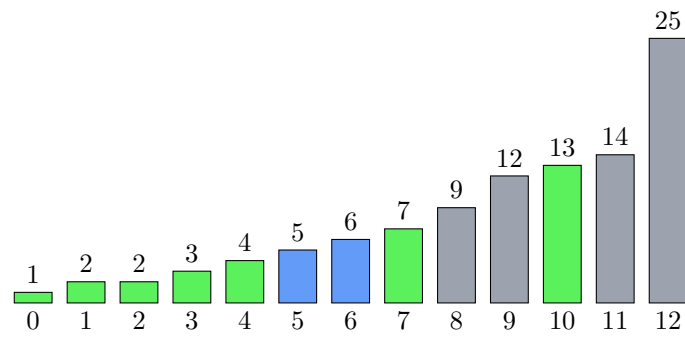


Quick Sort

Log

Using the last element of this section, value 6 at index 6, as the pivot.

Chart

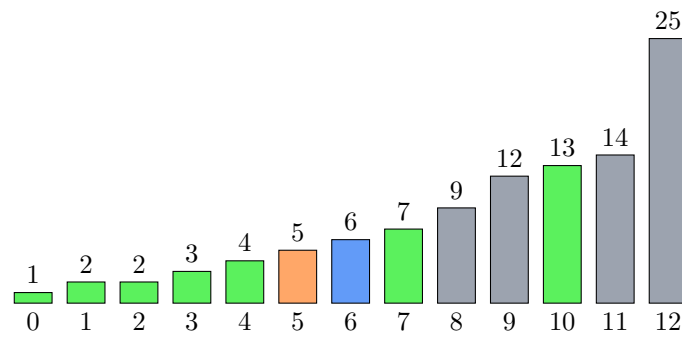


Quick Sort

Log

Checking whether the element at index 5 is smaller than the pivot (6).

Chart

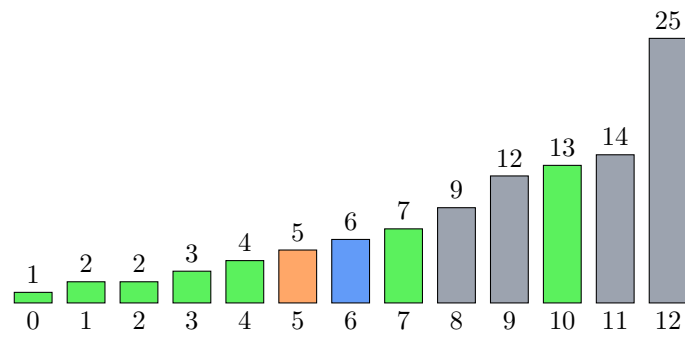


Quick Sort

Log

Element at index 5 is smaller than the pivot and already sits right after the smaller-than-pivot region

Chart

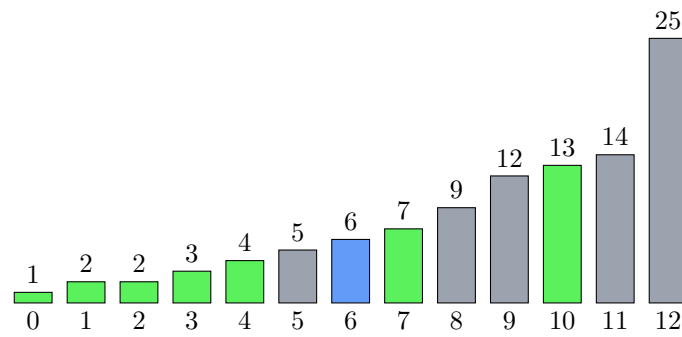


Quick Sort

Log

Element at index 5 is smaller than the pivot and already sits right after the smaller-than-pivot region

Chart

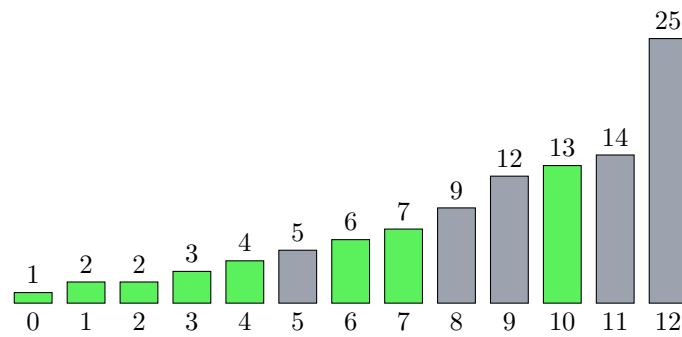


Quick Sort

Log

Swapping the pivot into index 6 | every element before it is now smaller, every element from there on i

Chart

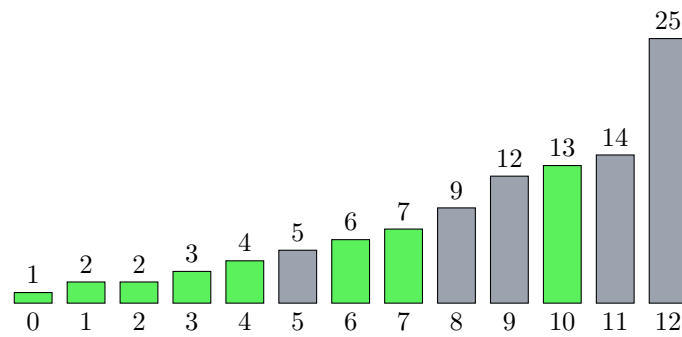


Quick Sort

Log

Swapping the pivot into index 6 | every element before it is now smaller, every element from there on i

Chart

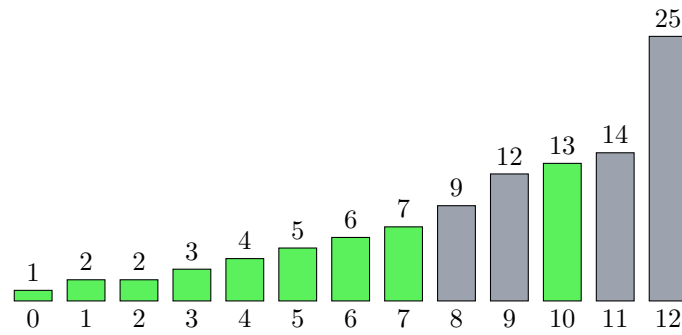


Quick Sort

Log

The section at index 5 contains only one element, so it's already in its final position.

Chart

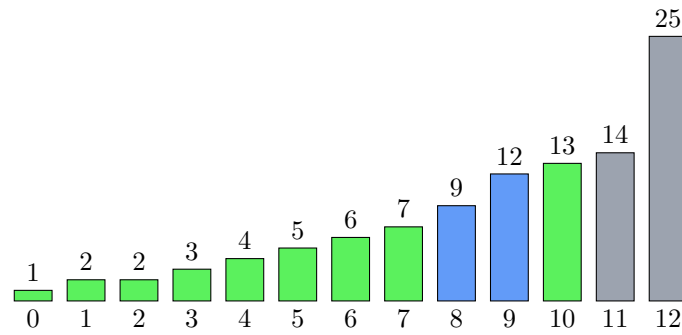


Quick Sort

Log

Partitioning and recursively sorting the section from index 8 to 9.

Chart

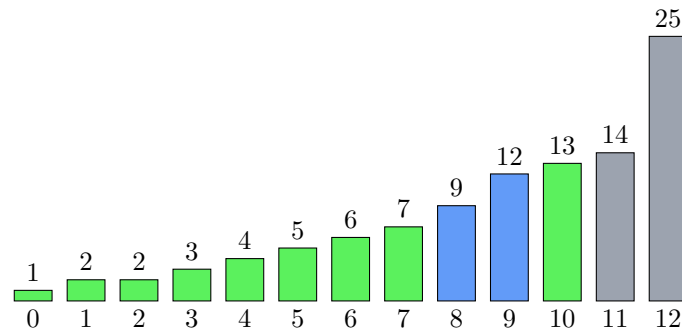


Quick Sort

Log

Using the last element of this section, value 12 at index 9, as the pivot.

Chart

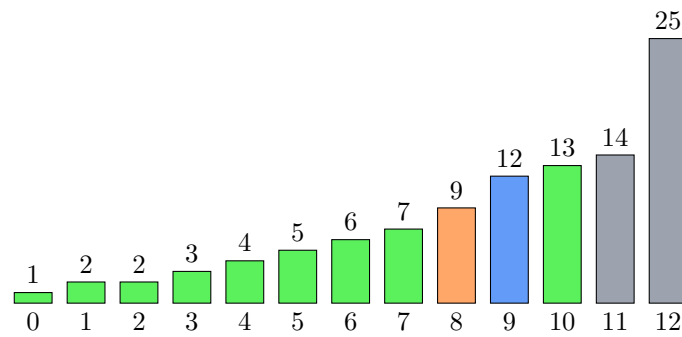


Quick Sort

Log

Checking whether the element at index 8 is smaller than the pivot (12).

Chart

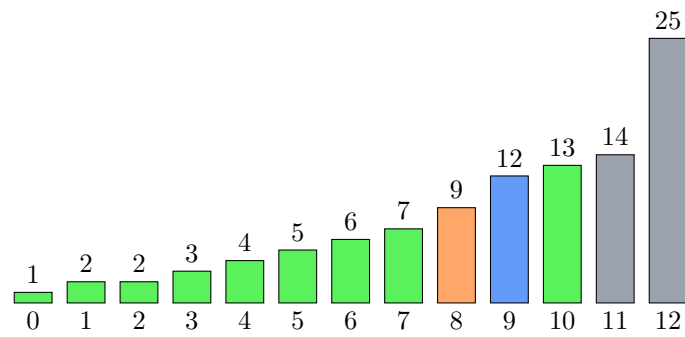


Quick Sort

Log

Element at index 8 is smaller than the pivot and already sits right after the smaller-than-pivot region

Chart

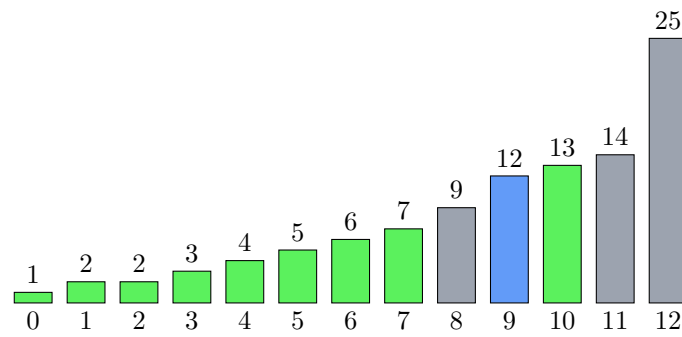


Quick Sort

Log

Element at index 8 is smaller than the pivot and already sits right after the smaller-than-pivot region

Chart

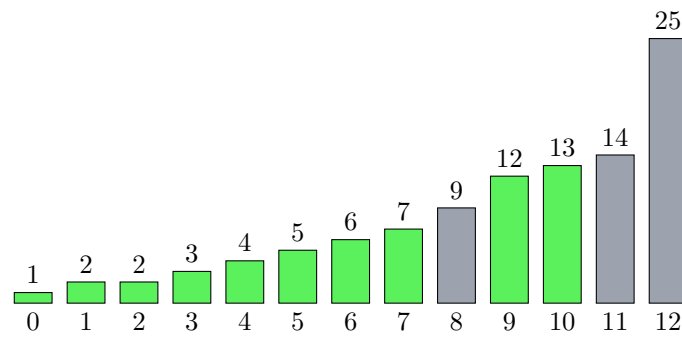


Quick Sort

Log

Swapping the pivot into index 9 | every element before it is now smaller, every element from there on i

Chart

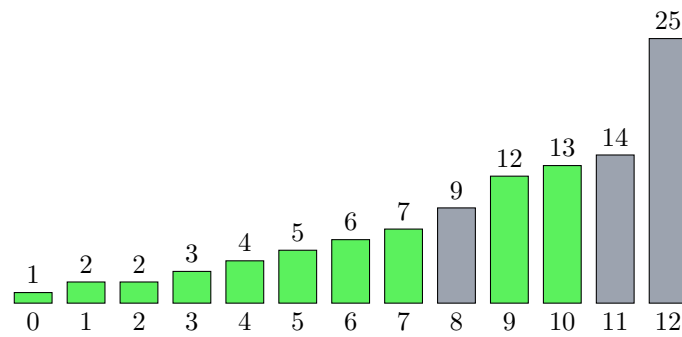


Quick Sort

Log

Swapping the pivot into index 9 | every element before it is now smaller, every element from there on i

Chart

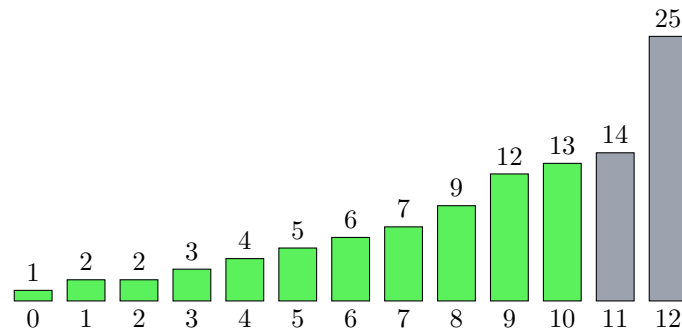


Quick Sort

Log

The section at index 8 contains only one element, so it's already in its final position.

Chart

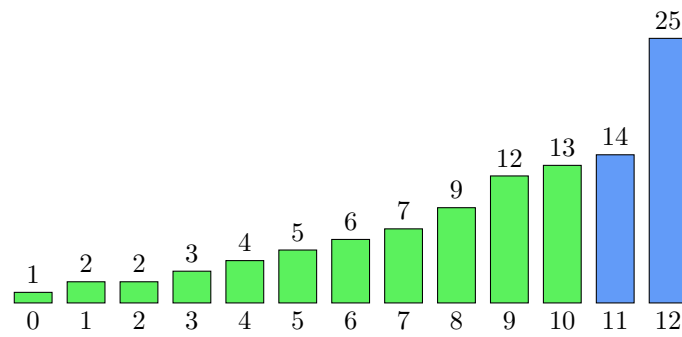


Quick Sort

Log

Partitioning and recursively sorting the section from index 11 to 12.

Chart

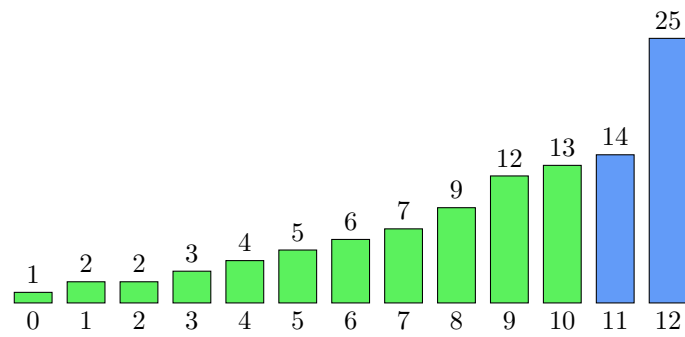


Quick Sort

Log

Using the last element of this section, value 25 at index 12, as the pivot.

Chart

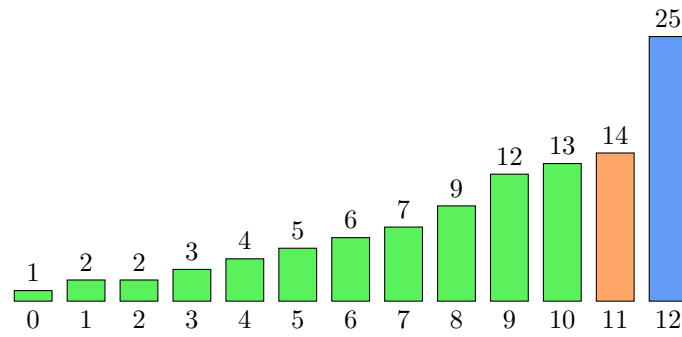


Quick Sort

Log

Checking whether the element at index 11 is smaller than the pivot (25).

Chart

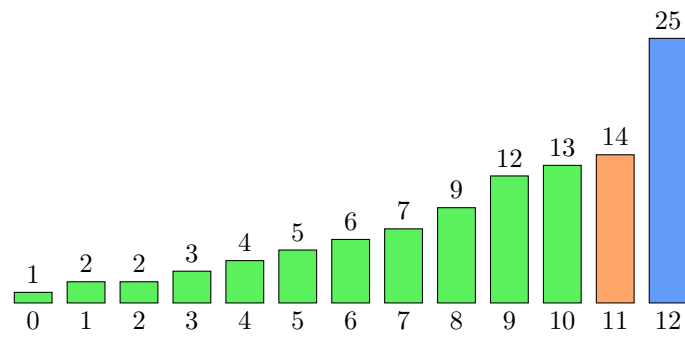


Quick Sort

Log

Element at index 11 is smaller than the pivot and already sits right after the smaller-than-pivot region

Chart

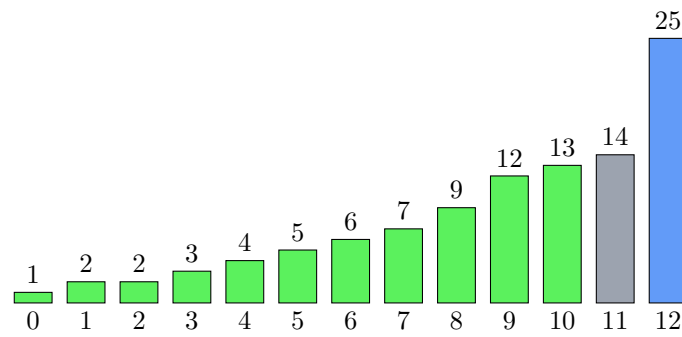


Quick Sort

Log

Element at index 11 is smaller than the pivot and already sits right after the smaller-than-pivot region

Chart

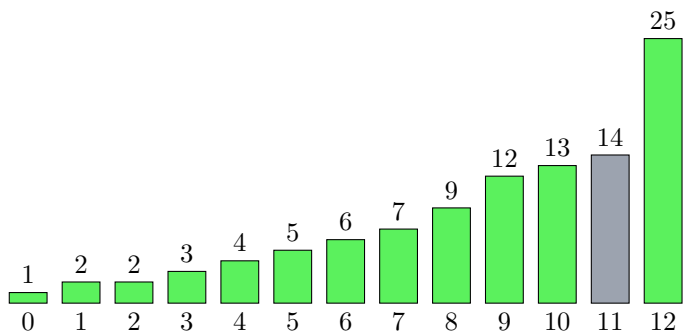


Quick Sort

Log

Swapping the pivot into index 12 | every element before it is now smaller, every element from there on

Chart

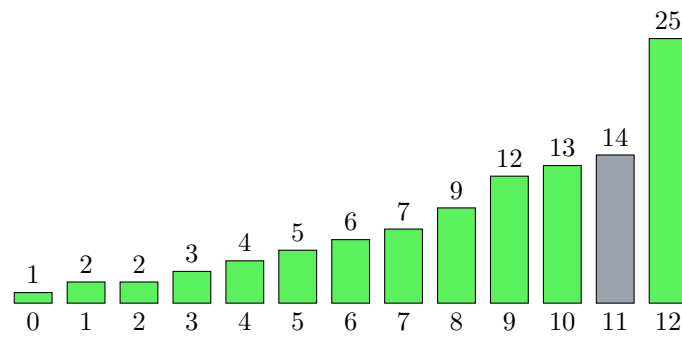


Quick Sort

Log

Swapping the pivot into index 12 | every element before it is now smaller, every element from there on

Chart

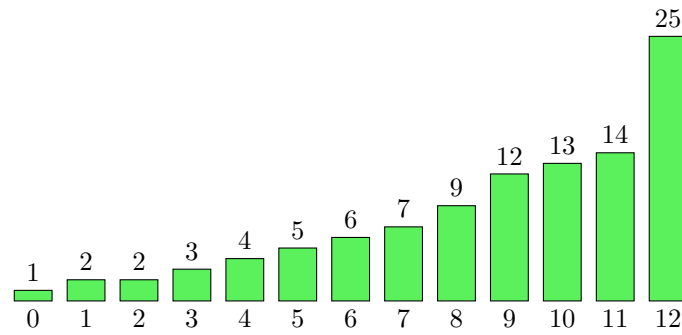


Quick Sort

Log

The section at index 11 contains only one element, so it's already in its final position.

Chart



Quick Sort

Log

Every section has been partitioned around its pivot | the array is fully sorted.

Chart

